An LM-type Unit Root Test for Functional Time Series

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Abstract

In this paper, we propose an Lagrange multiplier (LM)-type unit root test for functional time series. We derive the limit distribution of our proposed test statistics under the null hypothesis of a random walk and its asymptotic behavior of alternative hypotheses of trend stationary, weakly dependent stationary, and autoregressive stationary models. Specifically, we show the test consistency under all the aforementioned alternative hypotheses. Simulation studies are conducted to justify the theories and the desirable finite-sample performance of the proposed functional unit root test. The proposed test is also applied to real data of intraday stock price curves and the test results are plausible.

Keywords: Unit root test, Functional time series, Non-stationarity, Asymptotic statistics, Intraday stock price analysis

1. Introduction

Functional data analysis (FDA) has been popular in the last two decades and the conventional analytical tools and statistical inference are documented in Ramsay and Silverman (2005); Horváth and Kokoszka (2012). FDA is comparable with high-dimensional data analysis as the data frequently observed in certain space dimension could be viewed as functional data. Incorporated with the time dimension, functional time series (FTS) analysis attracts lots

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of attentions of statisticians and econometricians because of its wide applicability; see Hörmann and Kokoszka (2012). For example, we can find successful applications of FTS analysis for medical data for specific ages (see Erbas et al. (2007)), intraday electricity price curves (see Liebl (2013)), and seasonal environment data (see Cai et al. (2018)). In the context of econometrics, more and more refined concepts and models of FTS are proposed; see Horváth and Kokoszka (2012, Chapter 13) for a systematic review of functional autoregressive (FAR) models based largely on the monograph of Bosq (2000), Müller et al. (2011) for FDA for volatility, and Cerovecki et al. (2019) for functional GARCH models.

Most of the aforementioned studies are underlain by the stationarity assumption for the FTS data, which facilitates the projections of FTS on the basis system extracted from the principal components and thus allows us to leverage some scalar time series techniques to analyze FTS. A natural question is then how to test the stationarity or verify the model for the FTS data. The stationarity tests in Horváth et al. (2014) (in the time domain) and Aue and van Delft (2020) (in the frequency domain) and the trend stationarity test, namely Kwiatkowski–Phillips–Schmidt–Shin (KPSS) test, in Kokoszka and Young (2016) for FTS data are the well-established ones, while Chen and Pun (2019) provides a practical improvement of functional KPSS test by a bootstrap approach. Testing FTS for various assumptions has been a trending topic and to name a few, other examples include Horváth and Rice (2015) for the independence, Zhang (2016) for white noise testing and model diagnostic checking, Constantinou et al. (2017) for the separability of space-time domains, Górecki et al. (2018) for the normality, and Hörmann et al. (2018) for the periodicity. To construct a test for FTS, a limit distribution of the test statistic is usually derived under the null hypothesis, where the data are assumed to be stationary in most existing studies. To the best of our knowledge, the unit root test for FTS, where the null hypothesis presumes the non-stationarity of the data, is underexplored.

The unit root model has long been used in econometrics, including growth convergence theories, purchasing power parity hypothesis, and estimation of exchange rates and future interest rates. Testing for unit roots has been intensively studied since the seminal work of Dickey and Fuller (1979, 1981) for the scalar time series. Subsequently, Bhargava (1986); Dickey et al. (1986) develop the theory of unit root tests, especially their contribution to the deformation of unit root models, and Phillips and Perron (1988); Perron (1990) develops the Phillips–Perron test by making non-parametric correction to the

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statistic of the Dickey–Fuller test. For the pioneering scalar unit root test criteria above, a comparison is documented in Pantula et al. (1994). In the last two decades, the studies of unit root tests were extended to panel data. For example, Maddala and Wu (1999) conducts a comparative study of unit root tests under different null and alternative hypotheses; Levin et al. (2002) provides a systematic theory of panel unit root tests; Jansson and Moreira (2006) establishes power envelopes for testing regression models with nearly integrated regressors; Moon and Perron (2012) uses the false discovery rate in classification between unit root and stationary models; Westerlund and Larsson (2015) analyzes the local asymptotic power of panel unit root tests. More recently, Zhang and Chan (2018) proposes a Portmanteau-type test for unit roots and Zhang et al. (2018) extends the study to high-dimensional unit root tests, whose statistic is constructed with the largest eigenvalue of the kernel of the integrated operator. Our work is to extend the sequence of studies on unit root tests to FTS data.

In the literature on unit root tests, the test statistics are usually based on the ordinary or generalized least squares estimate of the coefficient of an autoregressive model. For the FTS data, the autoregressive regression usually relies on functional principal component analysis (FPCA) that is specific for stationary data. For the statistical inference on non-stationarity hypothesis, it poses technical challenges to use FPCA. An exception is the analytical framework developed in Chang et al. (2016) that analyzes non-stationary FTS and identifies unit roots through FPCA. Their tests stem from the Beveridge–Nelson decomposition (thoroughly studied in Phillips and Solo (1992)) and the estimation of the unit root and stationarity subspaces, which form the whole Hilbert space for the functional data. The dimension of the unit root subspace is assumed to be finite (M), which allows the use of FPCA. The null hypothesis is about the value of M and the sequential tests are proposed with decreasing values of M from a pre-specified large value until the null hypothesis is rejected. Then the consistent tests derived in Chang et al. (2016) can determine the true value of M with asymptotically probability one, where the special case of $M = 0$ implies no unit root. The research gap however exists for the hypothesis of random walk (i.e. integration of order one, $I(1)$) FTS, which is heuristically equivalent to the case when $M \to \infty$ in Chang et al. (2016). The $I(1)$ representation of (co-integrated) processes is a crucial pre-requisite of the co-integration analysis or other data analyses; see Beare et al. (2017); Seo and Beare (2019) on the co-integrated linear processes in Hilbert space and Bayes Hilbert space.

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In contrast to the generalized concepts of unit roots in Chang et al. (2016), we are concerned about the null hypothesis of an explicit unit root (random walk) model for the FTS. To this end, we need to get rid of the reliance of FPCA.

This paper proposes an Lagrange multiplier (LM)-type functional unit root test (FURT) statistic based on the partial sum of estimated errors. Theoretically, we derive the limit distribution of the proposed statistic under the null hypothesis of integration of order one and the asymptotic behavior of the statistic under three different alternative hypotheses, namely trend stationary, weakly dependent stationary, and autoregressive stationary models. The test consistency is validated theoretically and numerically. We also apply the proposed FURT to intraday price curves of 50 randomly chosen US stocks. As a remark, the LM-type test statistics, initiated and developed in Silvey (1959); Breusch and Pagan (1980); Nabeya and Tanaka (1988); Kwiatkowski et al. (1992), are prevailing in FTS tests such as Horváth et al. (2014); Kokoszka and Young (2016); Chen and Pun (2019). We built an R package[1] that can implement all these LM-type FTS tests, including all numerical experiments in this paper.

To highlight the major mathematical contribution of this paper, we realize that most existing FTS tests leverage a core function convergence theorem, namely Theorem 1.1 in Berkes et al. (2013), to develop the asymptotic theories of the test statistics under stationary or non-stationary hypotheses. The key to applying the theorem is to include the demeaning of estimated errors such that the mean will dominate the limit distribution of the statistic. However, when such a construction is extended to testing for unit root model, the partial sum of estimated errors becomes the difference between the last and first samples of FTS, which is not dominated by a mean term. Thus, Theorem 1.1 in Berkes et al. (2013) becomes unavailable. In this paper, we manage to get rid of the reliance of the theorem and show the test validity based on some fundamental limit theories.

The remainder of this paper is organized as follows. Section 2 states the problem and the proposed statistics of the FURT. Section 3 presents the limit distribution of the test statistics under the null hypothesis and the asymptotic behavior of the recommended test statistic under alternative hypotheses. Then, we conduct simulation studies in Section 4 and an empirical study of

[1]We will cite the package after the review.

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intraday stock price curves in Section 5. Finally, Section 6 concludes.

2. Problem Formulation and Test Statistics

2.1. The Problem

Suppose that N time series data are available in functional form (i.e. each is a curve), denoted by $\mathcal{X}_N = \{X_n(t)\}_{n=1}^N$, where the index t is the argument of the curves. Without loss of generality, we assume that the curves are being normalized and defined on the unit interval $t \in [0,1]$. Throughout this paper, all the random functions and functional parameters are defined on a Hilbert space $L^2([0,1])$ of all square-integrable real-valued functions on $[0,1]$ with an inner product $\langle f,g\rangle \triangleq \int_0^1 f(t)g(t)dt$ for $f,g \in L^2([0,1])$. To ease the notational burden, we suppress the limits of the integral, i.e. $\int f(t)dt \triangleq \int_0^1 f(t)dt$. We study the problem of our interest on a probability space $(\Omega, \mathcal{F}, \Pr)$, on which we define $\mathrm{E}(\cdot)$ as the expectation operator.

With the definition of integrated FTS, we consider the null hypothesis of FTS being integration of order one ($I(1)$) as follows:

Unit root non-stationary:

$$\begin{aligned} & H_0^{ur}: \ X_n(t) = \mu(t) + \sum_{i=1}^{n} \eta_i(t) \\ \text{or equivalently} \quad & H_0^{ur}: \ X_n(t) = X_{n-1}(t) + \eta_n(t) \end{aligned} \tag{1}$$

for $t \in [0,1],\ n = 1,\ldots,N$, where the intercept $\mu(t) \triangleq X_0(t)$ is a deterministic functional parameter and the errors $\{\eta_n\}_{n=1}^N$ are a subsequence of Bernoulli shifts $\{\eta_n\}_{n=-\infty}^{\infty}$, which are defined below.

Definition 1 (Benoulli shifts)**.** $\{\eta_n\}_{n=-\infty}^{\infty}$ forms a Bernoulli shifts if $\eta_n = f(\varepsilon_n, \varepsilon_{n-1}, \ldots)$ for some measurable function $f: \Omega^{\infty} \mapsto L^2$ and independent and identically distributed (i.i.d.) functions $\varepsilon_n, -\infty < n < \infty$. Moreover, $\mathrm{E}\{\eta_0(t)\} = 0$ for all t and $\mathrm{E}(\|\eta_0\|^{2+\delta}) < \infty$ for some $0 < \delta < 1$, and the sequence $\{\eta_n\}_{n=-\infty}^{\infty}$ can be approximated by m-dependent sequences $\{\eta_{n,l}\}_{n=-\infty}^{\infty}$ such that $\sum_{m=1}^{\infty}(\mathrm{E}\|\eta_n - \eta_{n,m}\|^{2+\delta})^{1/\kappa} < \infty$ for some $\kappa > 2+\delta$, where $\eta_{n,m}$ is defined by $\eta_{n,m} = f(\varepsilon_n, \ldots, \varepsilon_{n-m+1}, \varepsilon^*_{n-m}, \varepsilon^*_{n-m-1}, \ldots)$ with the ε^*'s being the independent copies of ε_0, and independent of any $\varepsilon_n, \forall n$.

In addition to the specifications of Benoulli shifts, we also assume that

$$\lim_{m\to\infty} m\mathrm{E}\|\eta_n - \eta_{n,m}\| \to 0, \tag{2}$$

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which is a very natural assumption following Horváth et al. (2013). All the μ and $\{\eta_n\}_{n=1}^N$ presented in this paper are assumed to be Bernoulli shifts with (2) if there is no further notification. With the errors process defined above, our framework is general enough to nest most stationary time series models in practice; see Wu (2005); Hörmann and Kokoszka (2010) for the investigation of weak dependence and Pötscher and Prucha (1997) for its applications in econometrics.

For testing the unit root model, a general alternative hypothesis may be that H_0^{ur} does not hold. However, it is common to consider the following specific alternative hypotheses to reveal practical insights. Note that the consideration of the alternative hypothesis does not influence the construction of our test statistic and its limit distribution but the testing rule will be determined accordingly to achieve a desirable power. The alternative hypotheses we consider are listed as follows:

Trend stationary:

$$H_1^{ts}: \ X_n(t) = \mu(t) + n\xi(t) + \eta_n(t) \tag{3}$$

for $t \in [0,1],\ n = 1, \ldots, N$, which is considered in Kokoszka and Young (2016), where the slope $\xi \in L^2([0,1])$ is a deterministic functional parameter. To be distinguished from the weakly dependent stationary model, we impose that $\|\xi\| > 0$.

Weakly dependent stationary:

$$H_1^{ws}: \ X_n(t) = \mu(t) + \eta_n(t) \tag{4}$$

for $t \in [0,1],\ n = 1, \ldots, N$, which is considered in Horváth et al. (2014). Under this alternative hypothesis, we impose that the difference of the errors $\Delta\eta_n(t) = \eta_n(t) - \eta_{n-1}(t), n = 2, \ldots, N$ cannot be part of (another) Bernoulli shifts, e.g., i.i.d. or moving average errors are not allowed. This is to avoid the situation that (4) may be indifferent from the unit root non-stationary model H_0^{ur}.

Autoregressive (of order 1, i.e. AR(1)) stationary:

$$H_1^{ar1}: \ X_n(t) - \mu(t) = \Psi\{X_{n-1}(t) - \mu(t)\} + \eta_n(t) \tag{5}$$

for $t \in [0,1],\ n = 1, \ldots, N$, which is considered in Horváth et al. (2010), where Ψ is an integral operator with the kernel function $\psi(s,t)$ satisfying $\iint \psi^2(s,t)dsdt < 1$.

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We remark that for the hypotheses above, the μ and ψ are unknown and the μ's are estimated differently according to the corresponding hypothesis.

2.2. An LM-type Test Statistic for Functional Unit Root Models

To motivate the proposal of our test statistic, we refer to a univariate version of the problem of our interest, i.e. random walk test with the null hypothesis $H_0^{ur} : X_n = \mu + \sum_{i=1}^{n} \varepsilon_i$, where the intercept $\mu = X_0$ is a deterministic parameter and $\{\varepsilon_n\}_{n=1}^{N}$ are assumed to be stationary errors with zero mean. The observation of this preliminary study assists us to mimic the extension from the KPSS test (Kwiatkowski et al. (1992)) to the functional stationarity test (Horváth et al. (2014)) or functional KPSS test (Kokoszka and Young (2016)), where the authors propose to use functional version of the numerator of the corresponding univariate LM test as their test statistics.

Analogous to Nabeya and Tanaka (1988); Kwiatkowski et al. (1992), we can derive an LM test statistic of the univariate random walk test, where the derivation can be found in Appendix A. In summary, the LM test statistic for univariate unit root model (random walk) is $LM = X^\top X/(X^\top A_N X)$, where $X = (X_1, \ldots, X_N)^\top$ and $A_N = (a_{ij})_{i,j=1,\ldots,N}$ with $a_{ij} = \min(i, j)$. Moreover, we reject the null hypothesis when LM is small. Analogous to Horváth et al. (2014), since the information of the denominator of LM is contained in its numerator and the long-run variance of the denominator is much more difficult to estimate compared to the numerator, we use only (the functional version of) the numerator of LM, i.e. $LM_{num} = N^{-2}X^\top X$ as our test statistic. However, this form is not easy for us to extend to its functional version and to derive its asymptotic theory. We rewrite it in terms of estimated errors as follows. Without loss of generality, we set $\mu = X_0 = 0$. The errors are estimated by $e_n = X_n - X_{n-1}$ and thus the LM_{num} can be rewritten as $LM_{num} = N^{-1}\sum_{n=1}^{N}\left(N^{-1/2}\sum_{i=1}^{n} e_i\right)^2$. The LM_{num}'s for different nonzero μ's are equivalent in the asymptotic sense.

Now, we are ready to introduce the functional version of LM_{num} as our test statistic. Given a set of observed FTS data $\{X_n(t)\}_{n=1}^{N}$, we consider the estimated errors under the null hypothesis H_0^{ur}

$$\hat{\eta}_n(t) \triangleq X_{n+1}(t) - X_n(t) \tag{6}$$

for $n \in \{1, 2, \ldots, N-1\}$. Then a suitable functional unit root test (FURT)

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statistic is defined by

$$R_N \triangleq \iint Z_N^2(x,t)dtdx, \quad \text{where } Z_N(x,t) \triangleq \frac{1}{\sqrt{N}} \sum_{n=1}^{\lfloor Nx \rfloor} \hat{\eta}_n(t). \tag{7}$$

Note that Z_N is the partial sum of the estimated errors. The similar LM-type test statistics are previously considered in some FTS tests; see Horváth et al. (2014); Kokoszka and Young (2016). Regarding the R_N, two points should be paid attention:

1. The $\hat{\eta}_n(t)$ in (6) estimates $\eta_{n-1}(t)$ in (1) and we do not estimate $\eta_1(t)$ in (1). Noting that we are developing an asymptotic test, a single sample of $\eta_1(t)$ does not affect the property of our proposed test and ignoring it gets rid of the test dependence on the intercept $\mu(t)$, which is the initial starting point of the FTS data or $X_0(t)$.
2. The $Z_N(x,t)$ in (7) contains $\hat{\eta}_{\lfloor Nx \rfloor}(t)$ and thus seemingly $X_{N+1}(t)$, which is not in the observed FTS data. In fact, the test statistic R_N in (7) can be rewritten as

$$R_N = \frac{1}{N} \iint \{X_{\lfloor Nx \rfloor+1}(t) - X_1(t)\}^2 dtdx = \frac{1}{N^2} \sum_{n=2}^{N} \int \{X_n(t) - X_1(t)\}^2 dt, \tag{8}$$

   from which, we can see that the evaluation of the test statistic R_N does not involve $X_{N+1}(t)$. Despite of the simpler expression of R_N in (8), we still opt for the expression in (7) as it facilitates later asymptotic analyses and provides more intuitive use of some existing results.

It is helpful to understand the differences between our FURT on the original series and the functional stationarity tests (FSTs) in Horváth et al. (2014) on the difference series. The primary difference is that the test consistency shown in Horváth et al. (2014) is no longer applicable as the alternative hypotheses are amended with the difference data. The secondary difference is that the weakly stationary model in Horváth et al. (2014) contains a drift term and thus the H_0^{ws} on the difference data is a more general unit root model than H_0^{ur}: $X_n(t) = X_{n-1}(t) + \xi(t) + \eta_n(t)$. Hence, FSTs has an additional demeaning step in their construction of test statistics. In asymptotics, we anticipate to estimate a zero $\xi(t)$ under H_0^{ur}. Therefore, FSTs on difference data may also be valid tests for H_0^{ur} but due to the primary difference, it is ambiguous about the asymptotic behavior of the FST statistics under

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the amended alternative hypotheses. Since FST is not our main focus (theoretically), we only compare our FURT and FSTs (on difference series) in simulations.

3. Asymptotic Theory of the Functional Unit Root Test

In this section, we first observe the asymptotic theory of the univariate version of FURT. Then, we derive the limit distributions of the test statistics R_N in (7) under the unit root non-stationary null hypothesis H_0^{ur}. Finally, we investigate the asymptotic behavior of R_N under alternative hypotheses. Throughout the paper, $W_1, W_2, \cdots$ are denoted as independent Wiener processes on $[0, 1]$. All the proofs in this section are deferred to Appendix B.

3.1. Asymptotic Theory of the Univariate LM-type Unit Root Tests

By applying many useful results in Kwiatkowski et al. (1992) to the univariate unit root test, we can easily obtain that under H_0^{ur},

$$LM_{num} \xrightarrow{d} \sigma^2 \int_0^1 W^2(r)dr \quad \text{as } N \to \infty,$$

where $\sigma^2 = \lim_{N\to\infty} N^{-1}\mathrm{E}(S_N^2)$ with $S_N = \sum_{n=1}^N e_n$ is the long-run variance and $W(r)$ is Wiener process on [0,1]. To obtain the critical value for the test with LM_{num}, we can use a consistent estimator of σ^2 as in Phillips and Perron (1988), namely $s^2(l) = N^{-1}\sum_{n=1}^N e_n^2 + 2N^{-1}\sum_{s=1}^l w(s,l)\sum_{t=s+1}^N e_t e_{t-s}$, where l is the lag truncation parameter requiring $l \to \infty$ as $N \to \infty$ (e.g., $l = o(N^{1/2})$) $w(s,l)$ is an optional weighting function (e.g., the Bartlett window $w(s,l) = 1 - s/(l+1)$). Then, the critical value $\hat{d}$ of the test with LM_{num} can be obtained as the quantile of the estimated limit distribution $s^2(l)\int_0^1 W^2(r)dr$.

Now, we would like to gain some insights from the asymptotic behaviors of the univariate LM-type unit root tests under different alternative hypotheses. The alternative hypotheses under our considerations are still trend stationary $H_1^{ts}: X_n = \mu + n\xi + \varepsilon_n$, weakly dependent stationary $H_1^{ws}: X_n = \mu + \varepsilon_n$, and AR(1) stationary $H_1^{ar1}: X_n - \mu = \alpha(X_{n-1} - \mu) + \varepsilon_n$ with $0 < \alpha < 1$. By some tedious derivations and results in Kwiatkowski et al. (1992), we can obtain that $LM_{num} = \mathcal{O}_{\mathrm{Pr}}(N)$ under H_1^{ts} and $LM_{num} = \mathcal{O}_{\mathrm{Pr}}(1/N)$ under H_1^{ws} or H_1^{ar1}. Alternatively, the derivation can be deduced from our proofs of Theorems 2, 4, and 5 below as they are functional versions of these

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results. Again, with the tedious derivations omitted (but can be referred to Theorems 3, 6, and 7 below), we know that the consistent estimator of σ^2, $s^2(l) = \mathcal{O}_{\Pr}(1)$ under H_1^{ts}. On the other hand, under H_1^{ws} or H_1^{ar1}, the series data are stationary, so are the estimated residuals e_n with zero mean. Then the consistent estimator $s^2(l)$ will tend to a finite long-run variance of these stationary residuals when $N \to \infty$.

The analyses above show that under H_1^{ts}, $\Pr(LM_{num} > \hat{d}_{1-\alpha}) \to 1$, while under H_1^{ws} or H_1^{ar1}, $\Pr(LM_{num} < \hat{d}_{\alpha}) \to 1$, when $N \to \infty$. Hence, to test H_0^{ur}/H_1^{ts}, we need to conduct a right-tailed test, and to test H_0^{ur}/H_1^{ws} or H_0^{ur}/H_1^{ar1}, we need to conduct a left-tailed test.

3.2. Limit Distribution of the Test Statistics under the Null Hypothesis

Analogous to the scalar case studied in Lo (1991), the limit distribution of the statistic R_N under H_0^{ur} is not parameter-free. Following similar FTS tests in Horváth et al. (2014) and Kokoszka and Young (2016), we characterize the limit distribution in terms of the eigenvalues of the long-run covariance function of the errors. Then the asymptotic test is implemented by estimating the limit distribution with consistent estimates of the eigenvalues.

First, we introduce a necessary regularity condition on the long-run covariance function of the errors $\{\eta_n\}$:

$$c(t,s) = \mathrm{E}\{\eta_0(t)\eta_0(s)\} + \sum_{n=1}^{\infty}[\mathrm{E}\{\eta_0(t)\eta_n(s)\} + \mathrm{E}\{\eta_0(s)\eta_n(t)\}], \tag{9}$$

where $\{\eta_n(t)\}_{n=0}^{\infty}$ are the subsequence of the Bernoulli shifts $\{\eta_n(t)\}_{n=-\infty}^{\infty}$, which have the same distribution as the errors $\{\eta_n(t)\}_{n=1}^{N}$. By Lemma 2.2 in Berkes et al. (2013), the infinite sum in the definition of $c(t,s)$ is well-defined on $L^2([0,1]\times[0,1])$, i.e. $\iint c^2(t,s)dsdt < \infty$. By the Cauchy–Schwarz inequality and weak dependence in Definition 1,

$$|\mathrm{E}\{\eta_0(t)\eta_n(s)\}| = |\mathrm{E}[\eta_0(t)\{\eta_n(s)-\eta_{n,n}(s)\}]| \leq \{\mathrm{E}\eta_0^2(t)\}^{1/2}[\mathrm{E}\{\eta_0(t)-\eta_{0,n}(t)\}^2]^{1/2}.$$

Then again by the Cauchy–Schwarz inequality for $\int f(t)g(t)dt$,

$$\int c(t,t)dt = \sum_{n=-\infty}^{\infty}\int \mathrm{E}\{\eta_0(t)\eta_n(t)\}dt \leq \sum_{n=-\infty}^{\infty}\left\{\mathrm{E}\|\eta_0\|^2\mathrm{E}\|\eta_0-\eta_{0,n}\|^2\right\}^{1/2},$$

and thus $\int c(t,t)dt < \infty$, which will be used in the later analyses. Moreover, the bivariate function $c(t,s)$ is positive-definite and therefore there exist

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eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots \geq 0$ and the corresponding orthonormal eigenfunctions $\phi_1(t), \phi_2(t), \ldots,\ t \in [0,1]$ satisfying $\lambda_i \phi_i(t) = \int c(t,s)\phi_i(s)ds$ for $i \geq 1$. The eigen-decomposition of $c(t,s)$ plays a crucial role in many FTS tests, see Horváth et al. (2014); Kokoszka and Young (2016).

The following theorems show the limit distributions of the FURT statistic under $I(1)$ null hypothesis:

Theorem 1. *Under the null hypothesis H_0^{ur} (1), we have*

$$R_N \xrightarrow{d} R_\infty \triangleq \sum_{i=1}^{\infty} \lambda_i \int W_i^2(x)dx \quad \text{as } N \to \infty,$$

where $\lambda_1 \geq \lambda_2 \geq \cdots \geq 0$ are the eigenvalues of the long-run covariance function $c(t,s)$ in (9) and $\{W_i(x)\}_{i=1}^{\infty}$ are i.i.d. Wiener processes.

The proof of Theorem 1 is documented in the Appendix B.1. It is noteworthy that the limit distribution R_∞ depends on the Wiener processes instead of Brownian bridges mainly because the construction of our test statistic gets rid of the de-meaning step. From Theorem 9 in Appendix B.1, we also know that $\sum_{i=1}^{\infty} \lambda_i < \infty$ under null hypothesis, which implies that the test statistic tends to a finite distribution in probability one.

In implementation, we can use a kernel estimator $\hat{c}_N(t,s)$, e.g. proposed in Horváth et al. (2013), as a proxy of the long-run covariance function $c(t,x)$:

$$\hat{c}_N(t,s) = \hat{\gamma}_0(t,s) + \sum_{n=1}^{N-1} K\left\{\frac{n}{h(N)}\right\}\{\hat{\gamma}_n(t,s) + \hat{\gamma}_n(s,t)\}, \tag{10}$$

$$\hat{\gamma}_n(t,s) = \frac{1}{N}\sum_{j=n+1}^{N} \hat{\eta}_j(t)\hat{\eta}_{j-n}(s),$$

in which the kernel function K and the smoothing bandwidth $h(N)$ are required to satisfy the following conditions:

- $K(0) = 1$, $K(u) = 0$ for $u > c$, and K is continuous on $[0,c]$ with some $c > 0$;
- $h(N) \to \infty$ but $h(N)/N \to 0$ as $N \to \infty$.

Once the kernel K and bandwidth $h(N)$ are specified, we can approximate the limit distribution R_∞ in Theorem 1 with

$$\hat{R}_\infty^{(M)} \triangleq \sum_{i=1}^{M} \hat{\lambda}_i \int W_i^2(x)dx, \tag{11}$$

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where $\hat{\lambda}_1 \geq \hat{\lambda}_2 \geq \ldots \geq 0$ are the estimated eigenvalues from the kernel estimator $\hat{c}_{N-1}(t,s)$ in (10) and M is a large number. In this paper, we choose $M = N$ as N is considered as a large number in our study.

Under the null hypothesis H_0^{ur} (1), since $\hat{\eta}_i(t) = \eta_{i+1}(t)$, it is proved in Horváth et al. (2013, Section 5) that

$$\iint \{\hat{c}_N(t,s) - c(t,s)\}^2 dtds \xrightarrow{p} 0 \quad \text{as } N \to \infty.$$

Together with the result in Dunford and Schwartz (1988, p.1091), we conclude that $\hat{\lambda}_i$ and $\hat{R}_\infty^{(N)}$ are consistent estimates of λ_i and R_∞, respectively, under H_0^{ur}. Thus, the critical value of R_N can be deduced based on $\hat{R}_\infty^{(N)}$.

3.3. Asymptotic Behavior of the FURT under the Alternative Hypotheses

To validate an asymptotic test and to achieve a desirable testing power, it is crucial to understand the asymptotic behavior of the test statistic under the alternative hypothesis. In general, it is analytically challenging due to the inconsistency between the construction of the test statistic and the model under the alternative hypothesis. However, we manage to analyze our test statistic for three main classes of alternative hypotheses.

3.3.1. Alternative Hypothesis H_1^{ts}

For the alternative hypothesis of the trend stationary model H_1^{ts} (3), our analysis is thorough in the sense that a combination of Theorems 2 and 3 below shows the asymptotic power of one.

We first examine the asymptotic behavior of the test statistic under H_1^{ts}.

Theorem 2. *Under alternative hypothesis H_1^{ts} in* (3)*, we have*

$$\frac{1}{N} R_N \xrightarrow{\Pr} \|\xi\|^2 \quad \textit{as } N \to \infty.$$

By Theorem 2 and noting that $\|\xi\|^2 > 0$ is independent of N, we have that under H_1^{ts}, the test statistic R_N tends to infinite in probability in the order of N.

Next, we perform asymptotic analysis of the estimated limit distribution $\hat{R}_\infty$ under H_1^{ts}.

Theorem 3. *Under alternative hypothesis H_1^{ts} in* (3)*, we have*

$$\frac{1}{N} \hat{\lambda}_1 \xrightarrow{\Pr} 0 \quad \textit{as } N \to \infty,$$

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where $\hat{\lambda}_1$ *is the largest eigenvalue of the estimated long-run covariance function* $\hat{c}_N(t,s)$ *defined in* (10)*.*

Theorems 2 and 3 imply that under H_1^{ts}, the largest eigenvalue of estimated long-run covariance function $\hat{c}_N(t,s)$ tends to infinite always slower than R_N in probability. By the result of Dunford and Schwartz (1988, p.1091), we know that $\hat{R}_\infty^{(N)}$ and the critical value (lower-tailed α-quantile) of $\hat{R}_\infty^{(N)}$, denoted by $\hat{d}_{N,\alpha}$, evolve slower than R_N in N (in probability). In other words, we can conclude that under H_1^{ts},

$$\Pr(R_N > \hat{d}_{N,1-\alpha}) \to 1 \quad \text{as } N \to \infty,$$

which implies that the H_0^{ur}/H_1^{ts} right-tailed test is consistent.

3.3.2. Alternative Hypotheses H_1^{ws} *and* H_1^{ar1}

For the alternative hypotheses of the weakly dependent stationary model H_1^{ws} (4) and AR(1) stationary model H_1^{ar1} (5), we similarly use the combinations of Theorems 4 & 6 and 5 & 7 below to show the asymptotic power of one for our proposed FURTs.

We first examine the asymptotic behavior of the test statistic under H_1^{ws}.

Theorem 4. *Under alternative hypothesis* H_1^{ws} *in* (4)*, we have*

$$R_N = \mathcal{O}_{\Pr}(1/N).$$

Theorem 4 shows that the test statistic R_N tends to zero in probability in the order of $1/N$ under H_1^{ws}.

Under H_1^{ar1}, we need to specify the integral operator Ψ or its kernel ψ. The general ψ is difficult to study due to the intractable asymptotics of the $\hat{c}_N(t,s)$ under H_1^{ar1}. Therefore, we focus on a *constant kernel* function $\psi \equiv a$ for a constant $0 < a < 1$ and the analysis has been very complicated.

Theorem 5. *Under alternative hypothesis* H_1^{ar1} *in* (5) *with kernel function* $\psi(s,t) \equiv a$ *for* $0 < a < 1$*, we have*

$$R_N = \mathcal{O}_{\Pr}(1/N).$$

Theorem 5 shows that the test statistic R_N also tends to zero in probability in the order of $1/N$ under H_1^{ar1} with a constant kernel. We remark here that Theorem 5 is likely extendable to a piecewise constant kernel function

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by repeating the analysis in each piece. However, since it does not bring additional insights, we skip the possible extension such that this paper is presented within reasonable limits of space.

Next, we perform asymptotic analysis of the estimated limit distribution $\hat{R}_\infty$ under H_1^{ws} and H_1^{ar1}.

Theorem 6. *Under alternative hypothesis H_1^{ws} in (4), we have*

$$\hat{\lambda}_1 \xrightarrow{\Pr} K_{ws} \quad as\ N \to \infty,$$

where K_{ws} is a positive finite value.

Theorem 7. *Under alternative hypothesis H_1^{ar1} in (5) with kernel function $\psi(s,t) \equiv a$ for $0 < a < 1$, we have*

$$\hat{\lambda}_1 \xrightarrow{\Pr} K_{ar1} \quad as\ N \to \infty,$$

where K_{ar1} is a positive finite value.

Theorems 6 and 7 show that the largest eigenvalue of estimated long-run covariance function $\hat{c}_N(t,s)$ tends to a constant value. By the result of Dunford and Schwartz (1988, p.1091) and similar analysis as in the H_1^{ts} case, we can conclude that under H_1^{ws} or H_1^{ar1},

$$\Pr(R_N < \hat{d}_{N,\alpha}) \to 1 \quad \text{as } N \to \infty,$$

which implies that the H_0^{ur}/H_1^{ws} and H_0^{ur}/H_1^{ar1} left-tailed tests are consistent.

3.4. Summary of the Asymptotic FURTs

Theorems 2-7 reveal that the test statistic R_N in (7) behaves differently for different hypotheses and thus the decision rule has to be designed accordingly to achieve a desirable power. From the discussions in Section 3.3, we summarize our FURT with significance level α as follows:

- for H_1^{ts}, we reject H_0^{ur} if $R_N > \hat{d}_{N,1-\alpha}$;
- for H_1^{ws} or H_1^{ar1}, we reject H_0^{ur} if $R_N < \hat{d}_{N,\alpha}$,

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where $\hat{d}_{N,\alpha}$ is the lower-tailed α-quantile of $\hat{R}_{\infty}^{(N)}$ given in (11) and in implementation, it can be determined based on Monte Carlo simulations.

Before we end this section, we comment on the heuristic causes of different asymptotic behaviors under different hypotheses, which are not observed in functional stationarity test in Horváth et al. (2014) and functional KPSS test in Kokoszka and Young (2016). As the preceding papers and this paper consider a "partial sum of estimated errors"-based test statistic, its (limit) distribution is determined by the mean of the realized errors. No matter under which hypothesis, the errors play a dominant role in the determination of the distribution compared to other terms in the model. However, in the FURT, the partial sum of estimated errors, $Z_N(x,t)$ in (7), depends only on the realized error between the first and the last indexed sample data, regardless of the components other than the errors in the model. This feature dampens the impacts of the realized errors in the trend stationary model H_0^{ts} on the distribution of the test statistic compared to the non-zero trend $\xi(t)$. Thus, the trend term stands out and we can see from Theorem 2 that the growth rate of the test statistic is driven by the trend.

For the asymptotic analysis of R_N under H_1^{ws} or H_1^{ar1}, techniques with sort of a central limit theorem for FTS become infeasible since the test statistic is not dominated by the mean of the realized errors. It poses analytical challenges to derive the limit distribution. Therefore, we turn to study their asymptotic properties in Theorems 4 and 5 based on the fundamental features of the model. Since under H_1^{ws} or H_1^{ar1}, they do not have an accumulation (e.g. the slope in trend stationary model (3)) in the drift of their models. Heuristically, the test statistic R_N will be vanishing due to the averaging. Therefore, R_N behaves differently under different hypotheses.

4. Simulation Studies

In this section, we conduct simulation studies to examine the finite-sample performance of our proposed FURTs. Following the setting in similar FTS tests of Horváth et al. (2014); Kokoszka and Young (2016); Chen and Pun (2019), we choose the kernel function

$$K(t) = \begin{cases} 1, & \text{if } 0 \le |t| < 0.1, \\ 1.1 - |t|, & \text{if } 0.1 \le |t| < 1.1, \\ 0, & \text{if } |t| \ge 1.1. \end{cases}$$

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and examine two smoothing bandwidth functions $h = N^{2/5}$ and $h = N^{1/2}$, which used in Kokoszka and Young (2016), in the empirical size and power tests. After we provide the details of our simulations and implementations, we present the results of the FURTs with the statistic R_N, while the results of the FSTs with statistics T_N and M_N in Horváth et al. (2014) on the difference data are documented in Appendix C.

4.1. Details of Simulation and Implementation of the FURTs

To resemble the real-data analysis, we collect the empirical data of monthly US treasury yield curves from January 1970 to December 2002 in R package "fds," denoted by $\{X_n(t)\}_{n=1}^{396}$ (the curves are normalized and defined on $[0, 1]$). We first use these data to calibrate the model parameters under the respective hypothesis. Then, we simulate FTS data of length N, denoted by $\{\widetilde{X}_n(t)\}_{n=1}^{N}$, under different hypotheses with the calibrated model parameters. We consider different sample sizes: N =100 and 200 to examine the effects of sample size on our FURTs. For each hypothesis and sample size N, we repeat the simulation 1000 times to evaluate the testing performance. In all computations, we use 240 equidistant points to estimate every functional data, i.e. the curve values at $t_i = i/240,\ i = 1, \ldots, 240$.

To examine the empirical size and power of the FURT, we apply it to the simulated data, which are generated under the hypotheses H_0^{ur}, H_1^{ts}, H_1^{ws}, and H_1^{ar1} in (1) and (3)-(5), respectively. For the model parameter calibration, we simply estimate the mean function $\mu(t)$ by $\hat{\mu}(t) = X_1(t)$ for $t \in [0, 1]$ and estimate the errors with $\{\hat{\eta}_n(t)\}_{n=1}^{395}$ by (6) from the real data $\{X_n(t)\}_{n=1}^{396}$. We find the first two principle components $\hat{\phi}_1(t)$ and $\hat{\phi}_2(t)$ of the empirical covariance function of $\{\hat{\eta}_n(t)\}_{n=1}^{395}$, and then compute the standard errors of the first two functional scores of $\{\hat{\eta}_n(t)\}_{n=1}^{395}$, denoted by $\hat{\sigma}_1$ and $\hat{\sigma}_2$. The $\hat{\mu}(t)$ and the $\{\hat{\eta}_n(t)\}_{n=1}^{395}$ are then used in the simulations under different hypotheses.

In the simulations of different models, the $\mu(t)$ is replaced by $\hat{\mu}(t)$ and the errors $\{\widetilde{\eta}_n(t)\}_{n=1}^{N}$ are simulated as follows: $\widetilde{\eta}_n(t) = a_n\hat{\phi}_1(t) + b_n\hat{\phi}_2(t),\ t \in [0, 1],\ n = 1, \ldots, N$, where $\{a_n\}_{n=1}^{N}$ and $\{b_n\}_{n=1}^{N}$ are i.i.d. or AR(1) random sequences following normal distribution or Student's t-distribution with 5 degrees of freedom (t_5 distribution) with mean equal to 0 and variances equal to $\hat{\sigma}_1^2$ and $\hat{\sigma}_2^2$, respectively. Specifically,

- to generate t_5-distributed variables with a specified variance, we multiply the standard t_5-distributed random variables by a suitable constant

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$(\sqrt{3/5})$;

- to generate AR(1) sequences $\{a_n\}_{n=1}^N$, we set $a_1 = \epsilon_1$ and $a_n = \frac{1}{2}a_{n-1} + \frac{\sqrt{3}}{2}\epsilon_n$ for $n = 2, \ldots, N$, where $\epsilon_1, \epsilon_2, \ldots, \epsilon_N$ follow i.i.d. $N(0, \hat{\sigma}_1^2)$. The same applies to the generation of AR(1) sequences $\{b_n\}_{n=1}^N$.

Then, we are ready to simulate the data $\{\widetilde{X}_n(t)\}_{n=1}^N$ under different hypotheses:

- Under H_0^{ur} in (1), the simulated data are constructed as

$$\widetilde{X}_n^{ur}(t) \triangleq \hat{\mu}(t) + \sum_{i=1}^{n} \widetilde{\eta}_n(t), \quad t \in [0,1],\ n = 1, 2, \ldots, N.$$

- Under H_1^{ts} in (3), we simulate a Wiener process $\{W(t)\}_{t\in[0,1]}$ and set the slope $\widetilde{\xi}(t) \equiv 5W(t)$, where $W(t)$ is fixed throughout all simulations. The motivation behind this specification is to ensure that it is significant in the comparison with the errors and there is no specific pattern for the slope. Then, the simulated data under H_1^{ts} are constructed as

$$\widetilde{X}_n^{ts}(t) \triangleq \hat{\mu}(t) + n\widetilde{\xi}(t) + \widetilde{\eta}_n(t), \quad t \in [0,1],\ n = 1, 2, \cdots, N.$$

- Under H_1^{ws} in (4), the simulated data under H_1^{ws} are constructed as

$$\widetilde{X}_n^{ws}(t) \triangleq \hat{\mu}(t) + \widetilde{\eta}_n(t), \quad t \in [0,1],\ n = 1, \cdots, N.$$

- Under H_1^{ar1} in (5), the simulated data under H_1^{ar1} are constructed as $\widetilde{X}_1^{ar1}(t) = \hat{\mu}(t) + \widetilde{\eta}_1(t)$ and

$$\widetilde{X}_n^{ar1}(t) \triangleq \hat{\mu}(t) + \int \psi(t,s)\{\widetilde{X}_{n-1}^{ar1}(s) - \hat{\mu}(s)\}ds + \widetilde{\eta}_n(t)$$

for $t \in [0,1],\ n = 2, \cdots, N$, where we consider two kernel functions:

$$\psi_1(t,s) = 0.9 \quad \text{and} \quad \psi_2(t,s) = 0.3416 * \exp\left(\frac{t^2+s^2}{2}\right), \quad t,s \in [0,1],$$

in which ψ_1 is a *constant* kernel and ψ_2 is an *exponential* kernel with the Hilbert–Schmidt norm being approximately 0.5. The setting of the exponential kernel follows that in Horváth et al. (2014).

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For each set of the simulated FTS, we construct our FURT statistic in (7) or (8) with $\{\widetilde{X}_n(t)\}_{n=1}^N$. According to the decision rules in Section 3.4, we record the decision of rejection of not. Then for 1000 simulations, the rejection frequency for the simulated FTS under the null hypothesis will be an empirical size and that for the simulated FTS under alternative hypotheses will be an empirical power of FURT.

4.2. Simulation Results: Empirical Sizes and Powers of the FURTs

Tables 1 to 4 present different testing results under different hypotheses. The numbers in the tables are the rejection frequencies (in percentage) among 1000 simulations and they are interpreted as empirical size or power according to null or alternative hypothesis being considered, respectively. For each sample size N, we consider two significance levels $\alpha = 5\%$ and 10%. Since different decision rules are used for H_0^{ur}/H_1^{ts} test and H_0^{ur}/H_1^{ws} (or H_0^{ur}/H_1^{ar1}), the empirical sizes will be examined differently and thus the results for the former are presented in Tables 1 and 3 while the results for the latter are presented in Tables 2 and 4. Tables 1 and 2 use the bandwidth of $h = N^{2/5}$ while Tables 3 and 4 use the bandwidth of $h = N^{1/2}$.

Table 1: Empirical sizes (%) and powers (%) for (right-tailed) FURTs for H_0^{ur} versus H_1^{ts} with the bandwidth of $h = N^{2/5}$.

Error type	i.i.d. Normal				AR(1) Normal			
N	100		200		100		200	
α	5%	10%	5%	10%	5%	10%	5%	10%
H_0^{ur}	3.6	5.5	3.7	9.8	3.3	7.4	3.4	8.5
H_1^{ts}	99.4	100.0	100.0	100.0	99.1	100.0	100.0	100.0
Error type	i.i.d. Student's t_5				AR(1) Student's t_5			
N	100		200		100		200	
α	5%	10%	5%	10%	5%	10%	5%	10%
H_0^{ur}	4.2	10.1	3.0	7.4	4.0	8.8	3.6	7.8
H_1^{ts}	99.6	99.8	99.9	99.9	99.7	99.7	99.7	99.7

We examine the results for using the bandwidth function of $h = N^{2/5}$ and we first focus on the empirical sizes in the Tables 1 and 2. The deviation of the estimated asymptotic distribution of R_∞ from the empirical distribution of R_N can be characterized by the difference between the empirical size and the desired significance level. No matter which of the left-tailed and right-tailed

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Table 2: Empirical sizes (%) and powers (%) for (left-tailed) FURTs for H_0^{ur} versus H_1^{ws} or H_1^{ar1} with the bandwidth of $h = N^{2/5}$.

Error type	i.i.d. Normal				AR(1) Normal			
N	100		200		100		200	
α	5%	10%	5%	10%	5%	10%	5%	10%
H_0^{ur}	3.5	6.3	4.2	6.7	3.6	6.5	4.0	6.9
H_1^{ws}	////////////////		////////////////		59.0	75.3	66.8	85.5
H_1^{ar1} (const. kernel)	60.6	85.3	95.4	98.2	62.0	83.1	96.0	98.9
H_1^{ar1} (exp. kernel)	56.4	75.4	60.0	81.3	79.7	90.1	92.0	97.7
Error type	i.i.d. Student's t_5				AR(1) Student's t_5			
N	100		200		100		200	
α	5%	10%	5%	10%	5%	10%	5%	10%
H_0^{ur}	2.1	6.6	4.1	9.3	2.2	6.9	3.3	9.4
H_1^{ws}	////////////////		////////////////		66.9	80.8	75.6	85.0
H_1^{ar1} (const. kernel)	62.0	83.1	96.0	98.9	65.4	85.7	98.8	99.6
H_1^{ar1} (exp. kernel)	60.0	76.3	71.4	82.0	79.7	89.2	92.2	95.8

significance tests we consider, we can see that all the empirical sizes are close to the corresponding specified significance levels under different stochastic environments. It can also be observed that the empirical size will be generally closer to the significance level when N increases. It is noteworthy that in the right-tailed test (H_0^{ur}/H_1^{ts}), the empirical size converges faster to the significance level for the normal errors than the t_5 errors but in the left-tailed test (H_0^{ur}/H_1^{ws} (or H_0^{ur}/H_1^{ar1})), the comparison result is opposite, where the convergence is faster for the t_5 errors. It is related to the convergence of the distribution of the test statistic and it happens that different tails would have different convergence rates. With the observations above, the result of Theorem 1 is validated numerically.

Next, we examine the performance (empirical power) of the FURTs under different alternative hypotheses. It can be seen from Table 1 that the H_0^{ur}/H_1^{ts} test has very high values of power under different stochastic environments. They are mostly very close to 100% while all exceed 99%. As implied by our asymptotic analysis, a significant slope in the trend stationary model is the key distinction from the unit root model. Our FURT works very well for the H_0^{ur}/H_1^{ts} test. For the weakly dependent FTS, we exclude the studies with the i.i.d. errors since in this case, the H_0^{ur} nests H_1^{ws} and the formulation of the FURT is ill-posed. From Table 2, we can observe that

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the powers of the H_0^{ur}/H_1^{ws} test are lower than that of other tests but we can still obtain about 85% of empirical power when $N = 200$. It can be understood from the matter of fact that H_0^{ur} and H_1^{ws} are similar due to the general definition of the errors. To achieve a desirable power, one would need to have more data, i.e. a larger sample size N. For the AR(1) stationary FTS, two kernel functions are considered, where for the constant kernel, we pick the coefficient of 0.9 that is fairly close to 1, leading to a near unit root test. Again, the empirical powers are high, no matter which kernel function or which type of errors is considered. An interesting observation is that the H_0^{ur}/H_1^{ar1} test with the exponential kernel does not uniformly perform better than the test with the constant kernel, which may be counter-intuitive because the exponential kernel we consider possesses a norm of 0.5, much smaller than 0.9. We thus remark that the norm of the coefficient function of the AR model does not necessarily make a distinction between stationary and non-stationary models as it is an overall measure.

Table 3: Empirical sizes (%) and powers (%) for (right-tailed) FURTs for H_0^{ur} versus H_1^{ts} with the bandwidth of $h = N^{1/2}$.

Error type	i.i.d. Normal				AR(1) Normal			
N	100		200		100		200	
α	5%	10%	5%	10%	5%	10%	5%	10%
H_0^{ur}	2.6	7.0	3.2	8.5	2.3	6.9	3.1	8.5
H_1^{ts}	100.0	100.0	99.8	99.8	99.9	100.0	99.8	99.8
Error type	i.i.d. Student's t_5				AR(1) Student's t_5			
N	100		200		100		200	
α	5%	10%	5%	10%	5%	10%	5%	10%
H_0^{ur}	2.7	9.7	2.5	7.2	2.6	9.6	2.3	7.2
H_1^{ts}	100.0	100.0	99.9	100.0	99.9	100.0	99.9	99.9

Now, we examine the performance of the same FURT statistic R_N with the bandwidth function of $h = N^{1/2}$ in Tables 3 and 4. From the results of H_0^{ur}/H_1^{ts} test in Table 3, a similar conclusion can be obtained as in Table 1 that uses $h = N^{2/5}$ and the results in both tables are comparable.

However, in testing H_0^{ur}/H_1^{ws} or H_0^{ur}/H_1^{ar1}, the test performance of using $h = N^{1/2}$ is inferior to that of using $h = N^{2/5}$. The test results of H_0^{ur}/H_1^{ws} and H_0^{ur}/H_1^{ar1} with exponential kernel are even below our expectation when $h = N^{1/2}$, which implies a larger bandwidth or larger sample size is desired

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Table 4: Empirical sizes (%) and powers (%) for (left-tailed) FURTs for H_0^{ur} versus H_1^{ws} or H_1^{ar1} with the bandwidth of $h = N^{1/2}$.

Error type	i.i.d. Normal				AR(1) Normal			
N	100		200		100		200	
α	5%	10%	5%	10%	5%	10%	5%	10%
H_0^{ur}	1.5	5.7	1.9	5.8	1.5	6.2	2.0	6.4
H_1^{ws}	////////		////////		2.3	13.1	0.5	6.2
H_1^{ar1} (const. kernel)	36.8	66.3	64.5	93.8	49.7	76.5	91.9	97.6
H_1^{ar1} (exp. kernel)	0.5	4.4	0.3	6.3	19.3	49.1	24.7	57.3
Error type	i.i.d. Student's t_5				AR(1) Student's t_5			
N	100		200		100		200	
α	5%	10%	5%	10%	5%	10%	5%	10%
H_0^{ur}	1.3	4.6	2.8	7.1	1.6	5.0	2.9	7.7
H_1^{ws}	////////		////////		2.7	13.9	0.8	8.1
H_1^{ar1} (const. kernel)	38.0	67.9	85.8	95.1	49.8	77.1	91.8	98.4
H_1^{ar1} (exp. kernel)	0.7	5.4	1.2	9.0	20.6	54.1	30.9	61.5

for these two tests.

In a brief summary, the results in Tables 1 to 4 validate the asymptotic theory in Section 3 in the sense that the empirical sizes are desirable and the empirical powers are close to 1 when the bandwidth is carefully chosen. Our simulation studies examine different alternative hypotheses, stochastic environments, sample sizes, and significance levels. The plausible results show that our FURT is robust. Moreover, we conclude that it is better to use the bandwidth of $h = N^{2/5}$ in all cases while $h = N^{1/2}$ is also acceptable for H_0^{ur}/H_1^{ts} test only.

In comparison with the parallel results of FSTs on the difference data in Appendix C, our FURT performance is obviously superior in no matter achieving desirable empirical sizes or high powers. Among many FSTs with different settings, the overall best results are obtained by using the test statistic T_N and $h = N^{2/5}$. It should be noted that there are two mysteries about the implementation of FSTs on the difference data, namely the choice of $h(N)$ and which left- or right-tailed tests should be conducted for different alternative hypotheses. For the former, we still try only $h = N^{2/5}$ and $h = N^{1/2}$. For the latter, we conduct both left- and right-tailed tests, from which we can clearly see that one of them will lead to powers very close to zero, and then the better one is presented. We observe that when $h = N^{2/5}$,

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H_0/H_1^{ts} is a right-tailed test and the others are left-tailed tests, which are consistent with ours. However, when $h = N^{1/2}$, H_0/H_1^{ar1} (constant kernel) is a left-tailed test and the others are right-tailed tests. Moreover, the results with $h = N^{1/2}$ are clearly worse off and thus it is not a good choice, which implies that the asymptotic theory of FSTs on difference data should be evaluated. However, it is beyond the scope of this paper.

5. Application of FURT to Intraday Stock Price Curves

A leading example of functional unit root model is the intraday stock price curves. The analysis of the intraday price curves has attracted much attention since 2010. For example, Gabrys et al. (2010); Müller et al. (2011); Gabrys et al. (2013); Kokoszka and Reimherr (2013); Kokoszka et al. (2015) use appropriate methods to study the shapes of price or volatility curves. For functional FTS testing, Horváth et al. (2013) tested the stationarity of the intraday price curves and Kokoszka and Young (2017) tested the trend stationarity. However, from the results of these two tests, the null hypothesis of stationary or trend stationary is rejected for most of the intraday price curves if we do not make specific transformations on the original curves. Therefore, we apply our FURT on the intraday stock price curves and evaluate its empirical performance.

The data we use are minutely price data of 50 US stocks randomly chosen from the Bats (BYX) Exchange. The period of the data is from 2 Jan 2020 to 29 May 2020, resulting in $N = 103$ curves for each stock while some may only have 101 curves due to missing data. The data are accessible from a Russian website *finam.ru* or the corresponding author's website[2]. Since not all the intraday price data are available in every minute, we simply adopt B-spline approach to estimate and obtain 390 time points for every day and each stock. Then we apply our FURT on the fitted curves for each stock. Two types of FURTs, according to Section 3.4, are considered, namely the left-tailed test (for H_0^{ur}/H_1^{ws} or H_0^{ur}/H_1^{ar1}) and the right-tailed test (for H_0^{ur}/H_1^{ts}). The test results are presented in Figure 1 in terms of p-values.

From Figure 1, we can first see that the p-values are spread quite uniformly in $[0, 1]$. If we consider H_1^{ws} or H_1^{ar1} as an alternative, we will reject the null hypothesis for 1 stock at the significance level of 5% and 9 stocks

[2]The FTS data used in our empirical study can be downloaded from https://xxx (hidden for blind review)

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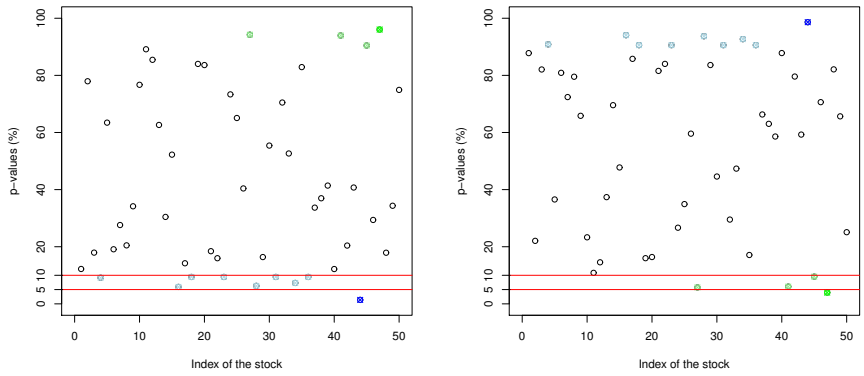


Figure 1: p-values of FURT of intraday price curves of 50 stocks. The left (resp.) panel presents the p-values of the left-tailed (resp. right-tailed) tests. The horizontal lines at levels of 5% and 10% are added to indicate that the point below the line suggest a rejection of FURT.

at the significance level of 10%, which are marked in blue and light blue, respectively in the left panel of Figure 1. If we consider H_1^{ts} as an alternative, we will reject the null hypothesis for 1 stock at the significance level of 5% and 4 stocks at the significance level of 10%, which are marked in green and light green, respectively in the right panel of Figure 1. Hence, there are 48 stocks (resp. 37 stocks), for which we cannot reject the null hypothesis of unit root model against any alternatives at the significance level of 5% (resp. 10%). The results are expected and aligned with our understanding of the financial market.

6. Conclusion

In this paper, we examine two FURT statistics with and without demeaning of the errors. We derive the limit distribution of the proposed test statistics under the $I(1)$ null hypothesis and investigate the asymptotic behavior of the non-demeaned FURT statistic under three alternative hypotheses. We show the test consistency theoretically. With the simulated data and real data, we validate the superior test performance of the non-demeaned FURT statistic R_N numerically and empirically.

There are some open problems remaining. It will be interesting to analyze

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the FURT under more other alternative hypotheses to bring more practical insights. However, the asymptotic theory, especially for the largest eigenvalue $\hat{\lambda}_1$ of the estimated long-run covariance function, is non-trivial. Moreover, the construction of the test statistic (in this paper, we used LM-type statistics) also has its critics that limit its applicability and ours is no exception. To derive the asymptotic results, this paper adopts an innovative approach, which is hopefully inspirational to the studies of non-stationary FTS tests.

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Appendix A. Derivation of the LM Test Statistic for Univariate Unit Root Test

The derivation below mainly follows Nabeya and Tanaka (1988); Kwiatkowski et al. (1992).

Let $\{X_n\}_{n=1}^N$ be an observable univariate time series. We assume that $\{X_n\}_{n=1}^N$ satisfy a general univariate model $X_n = \beta_n + \varepsilon_n$, where $\{\beta_n\}_{n=0}^N$ is a random walk, i.e. $\beta_n = \beta_{n-1} + u_n$ with $\{u_n\}_{n=1}^N$ being i.i.d. $N(0, \sigma_u^2)$ and β_0 being treated as a deterministic intercept, and $\{\varepsilon_n\}_{n=1}^N$ are i.i.d. $N(0, \sigma_\varepsilon^2)$. Note that a random walk model for $\{X_n\}_{n=1}^N$ is equivalent to having $\sigma_u > 0$ and $\sigma_\varepsilon = 0$ in this general model. Hence, to test the null hypothesis of (univariate) unit root model, we test $H_0 : \rho = \sigma_\varepsilon/\sigma_u = 0$ against $H_1 : \rho > 0$.

Note that $X_n = \beta_0 + \sum_{i=1}^n u_i + \varepsilon_n$. The observation vector $X = (X_1, \ldots, X_N)^\top$ is distributed as $N(\beta_0 \mathbf{1}_N, \sigma_u^2 A_N + \sigma_\varepsilon^2 I_N)$, where $\mathbf{1}_N = (1, \ldots, 1)^\top$, I_N is an N-dimensional identity matrix, and $A_N = (a_{ij})_{i,j=1,\ldots,N}$ with $a_{ij} = \min(i, j)$. In our random walk test, since the difference of data is the decisive factor instead of the intercept, the β_0 can be assumed to be 0 without loss of generality. Then we have $X \sim N(0, \sigma_u^2(\rho I_T + A_T))$ and the statistic $s(X) = X/\|X\|$ is a maximal invariant. Let $\Pr_\rho(\cdot)$ be the distribution of $s(X)$ and put

$$f_N(s(X)|\rho) = \frac{|A_N|^{1/2}}{|\rho I_N + A_N|^{1/2}} \left\{ \frac{X^\top(\rho I_N + A_N)X}{X^\top A_N X} \right\}^{-N/2},$$

which is the probability density of $\Pr_\rho(\cdot)$ with respect to $\Pr_0(\cdot)$. The rejection region of the LM test is now obtained as

$$\left. \frac{\partial \log f_N(s(X)|\rho)}{\partial \rho} \right|_{\rho=0} > constant.$$

Now, we simplify the statistic and the rule above as follows. Note that

$$\begin{aligned}
& \frac{\partial \log f_N(s(X)|\rho)}{\partial \rho} \\
= \; & \frac{\partial}{\partial \rho} \left[\frac{1}{2} \log |A_N| - \frac{1}{2} \log |\rho I_N + A_N| - \frac{N}{2} \log[X^\top(\rho I_N + A_N)X] + \frac{N}{2} \log(X^\top A_N X) \right] \\
= \; & -\frac{1}{2|\rho I_N + A_N|} \frac{\partial |\rho I_N + A_N|}{\partial \rho} - \frac{N}{2X^\top(\rho I_N + A_N)X} \frac{\partial X^\top(\rho I_N + A_N)X}{\partial \rho} \\
= \; & -\frac{1}{2|\rho I_N + A_N|} \frac{\partial |\rho I_N + A_N|}{\partial \rho} - \frac{N}{2X^\top(\rho I_N + A_N)X} X^\top X.
\end{aligned}$$

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When $\rho = 0$, we have $|\rho I_N + A_N| = |A_N| = 1$ and $\frac{\partial |\rho I_N + A_N|}{\partial \rho} = 2N - 1$. Hence, we obtain

$$\left.\frac{\partial \log f_N(s(X)|\rho)}{\partial \rho}\right|_{\rho=0} = -\frac{N}{2}\frac{X^\top X}{X^\top A_N X} - N + \frac{1}{2}.$$

The test statistic of the LM test can be simplified as $LM = N\frac{X^\top X}{X^\top A_N X}$ with rejection region $LM < constant$.

Appendix B. Proofs of Statements

Appendix B.1. Proof of Theorem 1

Before we present the proof of Theorem 1, we need some preliminary results. Theorem S.1 is a well-known theorem with the proof in Billingsley (1968). Theorem S.2 resembles Theorem 1.1 in Berkes et al. (2013), it is an important theorem to show the convergence of the sum of the eigenvalues of the long-run covariance function and the sum of the errors.

Theorem 8 (Theorem S.1.)**.** *Suppose Z_N, Y_N and Y are random variables taking values in a separable metric space with distance function ρ. If $Y_N \xrightarrow{d} Y$ and $\rho(Z_N, Y_N) \xrightarrow{p} 0$, then $Z_N \xrightarrow{d} Y$.*

Theorem 9 (Theorem S.2.)**.** *Under the null hypothesis H_0^{ur} in (1),*

1. *$\sum_{i=1}^{\infty} \lambda_i < \infty$, where $\lambda_1, \lambda_2, \ldots$ are eigenvalues of long-run covariance function $c(t,s)$ in (10).*
2. *We can define a sequence of Gaussian processes $\Gamma_N(x,t)$ such that:*

$$\{\Gamma_N(x,t), 0 \le x,t \le 1\} \overset{d}{=} \{\Gamma(x,t), 0 \le x,t \le 1\}$$

and

$$\kappa_N = \sup_{0 \le x \le 1} \|V_N(x,\cdot) - \Gamma_N(x,\cdot)\| = o_{\Pr}(1),$$

where $\Gamma(x,t) = \sum_{i=1}^{\infty} \lambda_i^{1/2} W_i(x)\phi_i(t)$ and $V_N(x,t) = \frac{1}{\sqrt{N}} \sum_{n=1}^{\lfloor Nx \rfloor} \eta_n(t)$.

Proof of Theorem 1. Noting that under H_0^{ur},

$$Z_N(x,t) = \frac{1}{\sqrt{N}} \sum_{n=1}^{\lfloor Nx \rfloor} \hat{\eta}_n(t) = \frac{1}{\sqrt{N}} \sum_{n=1}^{\lfloor Nx \rfloor} \{X_{n+1}(t) - X_n(t)\} = \frac{1}{\sqrt{N}} \sum_{n=1}^{\lfloor Nx \rfloor} \eta_n(t),$$

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we have

$$\sup_{0\le x\le 1} \|Z_N(x,\cdot) - \Gamma_N(x,\cdot)\| = o_{\Pr}(1)$$

from Theorem S.2. Hence, by Theorem S.1, we have

$$Z_N(x,t) \xrightarrow{d} \Gamma(x,t)$$

and by continuous mapping theorem, we thus have

$$R_N = \iint Z_N^2(x,t)dxdt \xrightarrow{d} \iint \{\Gamma(x,t)\}^2 dxdt = \sum_{i=1}^{\infty} \lambda_i \int W_i^2(x)dx,$$

which completes the proof of Theorem 1. □

Appendix B.2. Proof of Theorem 2

Proof of Theorem 2. Under alternative hypothesis H_1^{ts},

$$\begin{aligned} Z_N(x,t) &= \frac{1}{\sqrt{N}} \sum_{n=1}^{\lfloor Nx \rfloor} \{X_{n+1}(t) - X_n(t)\} \\ &= \frac{1}{\sqrt{N}} \sum_{n=1}^{\lfloor Nx \rfloor} \{\xi(t) + \eta_{n+1}(t) - \eta_n(t)\} \\ &= \frac{1}{\sqrt{N}} \{\lfloor Nx \rfloor \xi(t) + \eta_{\lfloor Nx \rfloor + 1}(t) - \eta_1(t)\}. \end{aligned}$$

Hence, we have

$$\begin{aligned} R_N &= \frac{1}{N} \iint \{\lfloor Nx \rfloor \xi(t) + \eta_{\lfloor Nx \rfloor + 1}(t) - \eta_1(t)\}^2 dxdt \\ &= \frac{1}{N^2} \sum_{n=1}^{N-1} \int \{n\xi(t) + \eta_{n+1}(t) - \eta_1(t)\}^2 dt \\ &= \frac{(N-1)N(2N-1)}{6N^2} \|\xi\|^2 + \frac{2}{N^2} \sum_{n=1}^{N-1} \int n\xi(t)\{\eta_{n+1}(t) - \eta_1(t)\} dt \\ &\quad + \frac{1}{N^2} \sum_{n=1}^{N-1} \int \{\eta_{n+1}(t) - \eta_1(t)\}^2 dt \\ &\triangleq R_1^* + R_2^* + R_3^*. \end{aligned}$$

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By the assumption of non-zero $\|\xi\|^2$, it is easy to see that R_1^* is of order N. Next, we evaluate the orders of R_2^* and R_3^*. For R_2^*, we note that

$$\begin{aligned}
&\left|\sum_{n=1}^{N-1}\int n\xi(t)\{\eta_{n+1}(t)-\eta_1(t)\}dt\right| \le \sum_{n=1}^{N-1} n\int |\xi(t)\{\eta_{n+1}(t)-\eta_1(t)\}|\,dt \\
\le\ & \frac{1}{2}\sum_{n=1}^{N-1} n\int[\xi^2(t)+\{\eta_{n+1}(t)-\eta_1(t)\}^2]dt \\
=\ & \frac{N(N-1)}{4}\|\xi\|^2+\frac{1}{2}\sum_{n=1}^{N-1} n\int\{\eta_{n+1}(t)-\eta_1(t)\}^2dt.
\end{aligned}$$

Then a direct calculation yields that

$$\begin{aligned}
\mathrm{E}\left[\sum_{n=1}^{N-1} n\int\{\eta_{n+1}(t)-\eta_1(t)\}^2dt\right] &\le \mathrm{E}\left[\sum_{n=1}^{N-1} 2n\int\{\eta_{n+1}^2(t)+\eta_1^2(t)\}dt\right] \\
&= 2N(N-1)\mathrm{E}\|\eta_0\|^2=\mathcal{O}(N^2).
\end{aligned}$$

Hence, we have $R_2^*=\mathcal{O}_{\mathrm{Pr}}(1)$.

Similarly, we have

$$\mathrm{E}(R_3^*)\le \mathrm{E}\left[\frac{2}{N^2}\sum_{n=1}^{N-1}\int\{\eta_{n+1}^2(t)+\eta_1^2(t)\}dt\right]=\frac{4(N-1)}{N^2}\mathrm{E}\|\eta_0\|^2=\mathcal{O}\left(\frac{1}{N}\right).$$

Therefore, we can conclude that under H_1^{ts}, we have

$$\frac{1}{N}R_N \xrightarrow{p} \frac{\|\xi\|^2}{3}, \quad \text{as } N\to\infty.$$

□

Appendix B.3. Proof of Theorem 3

Proof of Theorem 3. We mainly investigate the order of the estimate of the long-run covariance function $\hat{c}_N(t,s)$. Under the alternative hypothesis H_1^{ts}, we have

$$\hat{c}_N(t,s)=\hat{\gamma}_0(t,s)+\sum_{i=1}^{N-1}K\left\{\frac{i}{h(N)}\right\}\{\hat{\gamma}_i(t,s)+\hat{\gamma}_i(s,t)\} \tag{B.1}$$

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with

$$\begin{aligned}
\hat{\gamma}_i(t,s) &= \frac{1}{N}\sum_{j=i+1}^{N}\{X_{j+1}(t)-X_j(t)\}\{X_{j-i+1}(s)-X_{j-i}(s)\} \\
&= \frac{1}{N}\sum_{j=i+1}^{N}\{\xi(t)+\eta_{j+1}(t)-\eta_j(t)\}\{\xi(s)+\eta_{j-i+1}(s)-\eta_{j-i}(s)\}, \\
&= \frac{N-i}{N}\xi(t)\xi(s)+\frac{1}{N}\sum_{j=i+1}^{N}[\{\eta_{j+1}(t)-\eta_j(t)\}\{\eta_{j-i+1}(s)-\eta_{j-i}(s)\} \\
&\quad +\frac{1}{N}\sum_{j=i+1}^{N}[\xi(t)\{\eta_{j-i+1}(s)-\eta_{j-i}(s)\}+\{\eta_{j+1}(t)-\eta_j(t)\}\xi(s)]] \\
&\triangleq \hat{\gamma}_i^{(1)}(t,s)+\hat{\gamma}_i^{(2)}(t,s)+\hat{\gamma}_i^{(3)}(t,s)
\end{aligned}$$

for $i = 0, 1, \ldots, N-1$. By substituting different parts of $\hat{\gamma}_i^{(j)}$ for $i = 0, 1, \ldots, N-1$ into $\hat{c}_N(t,s)$, we denote it by $\hat{c}_N^{(j)}(t,s)$ for $j = 1, 2, 3$ and they compose $\hat{c}_N(t,s)$: $\hat{c}_N(t,s) = \hat{c}_N^{(1)}(t,s) + \hat{c}_N^{(2)}(t,s) + \hat{c}_N^{(3)}(t,s)$.

For the first term, it is easy to obtain that

$$\begin{aligned}
\hat{c}_N^{(1)}(t,s) &= \hat{\gamma}_0^{(1)}(t,s)+\sum_{i=1}^{N-1}K\left\{\frac{i}{h(N)}\right\}\left\{\hat{\gamma}_i^{(1)}(t,s)+\hat{\gamma}_i^{(1)}(s,t)\right\} \\
&= \xi(t)\xi(s)+\frac{1}{N}\sum_{i=1}^{N-1}2(N-i)K\left\{\frac{i}{h(N)}\right\}\xi(t)\xi(s).
\end{aligned}$$

Hence, we have

$$\begin{aligned}
\left\|\hat{c}_N^{(1)}\right\|^2 &= \left[1+\frac{1}{N}\sum_{i=0}^{N-1}2(N-i)K\left\{\frac{i}{h(N)}\right\}\right]^2\iint\xi^2(t)\xi^2(s)dtds \\
&\le \left[2\sum_{i=0}^{N-1}\left|K\left\{\frac{i}{h(N)}\right\}\right|\right]^2\|\xi\|^4 = \mathcal{O}\left[h^2(N)\left\{\int_0^\infty|K(u)|du\right\}^2\right] \\
&= \mathcal{O}\{h^2(N)\},
\end{aligned}$$

which implies that

$$\left\|\hat{c}_N^{(1)}\right\| = \mathcal{O}\{h(N)\}. \tag{B.2}$$

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For the second term of $\hat{c}_N^{(2)}(t,s)$, we first expand $\hat{\gamma}_i^{(2)}(t,s)$ into four terms:

$$\begin{aligned}
&\hat{\gamma}_i^{(2)}(t,s) \\
=\;& \frac{1}{N}\sum_{j=i+1}^{N}\{\eta_{j+1}(t)\eta_{j-i+1}(s)+\eta_j(t)\eta_{j-i}(s)-\eta_j(t)\eta_{j-i+1}(s)-\eta_{j+1}(t)\eta_{j-i}(s)\} \\
\triangleq\;& \hat{\gamma}_i^{(2,1)}(t,s)+\hat{\gamma}_i^{(2,2)}(t,s)-\hat{\gamma}_i^{(2,3)}(t,s)-\hat{\gamma}_i^{(2,4)}(t,s).
\end{aligned}$$

Analogously, we write $\hat{c}_N^{(2)}(t,s) \triangleq \hat{c}_N^{(2,1)}(t,s)+\hat{c}_N^{(2,2)}(t,s)+\hat{c}_N^{(2,3)}(t,s)+\hat{c}_N^{(2,4)}(t,s)$ where $\hat{c}_N^{(2,k)}(t,s)$ is equal to $\hat{c}_N(t,s)$ in (B.1) with $\hat{\gamma}_i$ replaced by $\hat{\gamma}_i^{(2,k)}$ for $k=1,2,3,4$. Then, we evaluate $\hat{c}_N^{(2,k)}(t,s)$ sequentially. For the $\hat{c}_N^{(2,1)}(t,s)$, by the ergodic theorem for random variables in the Hilbert space, we have

$$\iint\left[\hat{\gamma}_0(t,s)^{(2.1)}-\mathrm{E}\{\eta_0(t)\eta_0(s)\}\right]^2 dtds = o_{\mathrm{Pr}}(1).$$

Hence, as in Horváth et al. (2013, Appendix A), we obtain that

$$\left\|\sum_{i=1}^{N-1} K\left\{\frac{i}{h(N)}\right\}\hat{\gamma}_i^{(2,1)}-\tilde{c}\right\| = o_{\mathrm{Pr}}(1), \quad \forall\, t,s,$$

where $\tilde{c}(t,s)=\sum_{i\geq 1}\mathrm{E}\{\eta_0(s)\eta_i(t)\}$. Therefore, we have

$$\left\|\hat{c}_N^{(2,1)}-c\right\| = o_{\mathrm{Pr}}(1). \tag{B.3}$$

Similarly, we can also obtain that

$$\left\|\hat{c}_N^{(2,2)}-c\right\| = o_{\mathrm{Pr}}(1). \tag{B.4}$$

For $\hat{c}_N^{(2,3)}(t,s)$ and $\hat{c}_N^{(2,4)}(t,s)$, we can follow the similar approach in Horváth et al. (2013, Appendix A), where we only need to shift their index notation m to $m-1$ and $m+1$, respectively. Then we can obtain that

$$\iint\left[\hat{c}_N^{(2,3)}(t,s)-[c(t,s)+\mathrm{E}\{\eta_0(s)\eta_0(t)\}+\mathrm{E}\{\eta_0(s)\eta_1(t)\}]\right]^2 dtds = o_{\mathrm{Pr}}(1) \tag{B.5}$$

and

$$\iint\left[\hat{c}_N^{(2,4)}(t,s)-[c(t,s)-\mathrm{E}\{\eta_0(s)\eta_0(t)\}-\mathrm{E}\{\eta_0(s)\eta_1(t)\}]\right]^2 dtds = o_{\mathrm{Pr}}(1). \tag{B.6}$$

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By evaluations (B.3) to (B.6), we can conclude that

$$\left\|\hat{c}_N^{(2)}\right\| = o_{\Pr}(1). \tag{B.7}$$

For the third term of $\hat{c}_N^{(3)}(t,s)$, we can rewrite it as

$$\begin{aligned}
&\hat{c}_N^{(3)}(t,s) \\
=\ & \frac{1}{N}\xi(t)\left[\{\eta_{N+1}(s)-\eta_1(s)\}+\sum_{i=1}^{N-1}K\left\{\frac{i}{h(N)}\right\}\{\eta_{N+1}(s)+\eta_{N+1-i}(s)-\eta_{i+1}(s)-\eta_1(s)\}\right] \\
&+\frac{1}{N}\xi(s)\left[\{\eta_{N+1}(t)-\eta_1(t)\}+\sum_{i=1}^{N-1}K\left\{\frac{i}{h(N)}\right\}\{\eta_{N+1}(t)+\eta_{N+1-i}(t)-\eta_{i+1}(t)-\eta_1(t)\}\right] \\
\triangleq\ & \hat{c}_N^{(3,1)}(t,s)+\hat{c}_N^{(3,2)}(t,s).
\end{aligned}$$

It is easy to see that the evaluations on $\hat{c}_N^{(3,1)}(t,s)$and $\hat{c}_N^{(3,2)}(t,s)$ are the same. For the $\hat{c}_N^{(3,1)}(t,s)$, we note that by the Cauchy–Schwarz inequality,

$$\begin{aligned}
&\mathrm{E}\|\hat{c}_N^{(3,1)}\|^2 \\
\le\ & \frac{1}{N}\|\xi\|^2\left[\mathrm{E}\|\eta_{N+1}-\eta_1\|^2+\sum_{i=1}^{N-1}K^2\left\{\frac{i}{h(N)}\right\}\mathrm{E}\|\eta_{N+1}+\eta_{N+1-i}-\eta_{i+1}-\eta_1\|^2\right] \\
=\ & \mathcal{O}(1).
\end{aligned}$$

Hence, we can conclude that

$$\left\|\hat{c}_N^{(3)}\right\| = \mathcal{O}_{\Pr}(1). \tag{B.8}$$

Finally, by evaluations (B.2), (B.7), and (B.8), we can conclude that under the alternative hypothesis H_1^{ts}, we have

$$\|\hat{c}_N\| = \mathcal{O}_{\Pr}\{h(N)\},$$

which implies $\hat{\lambda}_1 = \mathcal{O}_{\Pr}\{h(N)\}$ by the result in Dunford and Schwartz (1988, p.1091). □

Appendix B.4. Proof of Theorem 4

Proof of Theorem 4. Under the alternative hypothesis H_1^{ws}, we have

$$\begin{aligned}
Z_N(x,t) &= \frac{1}{\sqrt{N}}\sum_{n=1}^{\lfloor Nx\rfloor}\{X_{n+1}(t)-X_n(t)\} = \frac{1}{\sqrt{N}}\sum_{n=1}^{\lfloor Nx\rfloor}\{\eta_{n+1}(t)-\eta_n(t)\} \\
&= \frac{1}{\sqrt{N}}\{\eta_{\lfloor Nx\rfloor+1}(t)-\eta_1(t)\}.
\end{aligned}$$

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Hence, under H_1^{ws},

$$\begin{aligned} R_N &= \iint Z_N^2(x,t)dxdt = \frac{1}{N}\iint \{\eta_{\lfloor Nx\rfloor+1}(t) - \eta_1(t)\}^2 dxdt \\ &= \frac{1}{N^2}\sum_{n=1}^{N-1}\int \{\eta_{n+1}(t) - \eta_1(t)\}^2 dt. \end{aligned}$$

Note that in this case,

$$\mathrm{E}(R_N) \le \mathrm{E}\left[\frac{2}{N^2}\sum_{n=1}^{N-1}\int \{\eta_{n+1}^2(t) + \eta_1^2(t)\}dt\right] = \frac{4(N-1)}{N^2}\mathrm{E}\|\eta_0\|^2 = \mathcal{O}\left(\frac{1}{N}\right).$$

Therefore, we have $R_N = \mathcal{O}_{\mathrm{Pr}}(1/N)$ under the alternative hypothesis H_1^{ws}. □

Appendix B.5. Proof of Theorem 5

Proof of Theorem 5. Under the alternative hypothesis H_1^{ar1}, we have

$$\begin{aligned} R_N &= \frac{1}{N}\iint \left\{\sum_{n=1}^{\lfloor Nx\rfloor+1} a^{\lfloor Nx\rfloor+1-n}\eta_n(t) - \eta_1(t)\right\}^2 dxdt \\ &= \frac{1}{N^2}\int \sum_{i=1}^{N}\left\{\sum_{n=1}^{i} a^{i-n}\eta_n(t) - \eta_1(t)\right\}^2 dt \\ &= \frac{1}{N^2}\int \left[\sum_{i=1}^{N}\left\{\sum_{n=1}^{i} a^{i-n}\eta_n(t)\right\}^2 + N\eta_1^2(t) - 2\eta_1(t)\sum_{i=1}^{N}\sum_{n=1}^{i} a^{i-n}\eta_n(t)\right] dt. \end{aligned}$$

Then we have

$$\begin{aligned} \mathrm{E}(R_N) &= \frac{1}{N}\int \mathrm{E}\{\eta_1^2(t)\}dt - \frac{2}{N^2}\int \mathrm{E}\left\{\eta_1(t)\sum_{i=1}^{N}\sum_{n=1}^{i} a^{i-n}\eta_n(t)\right\} dt \\ &\quad + \frac{1}{N^2}\int \mathrm{E}\left[\sum_{i=1}^{N}\left\{\sum_{n=1}^{i} a^{i-n}\eta_n(t)\right\}^2\right] dt. \end{aligned}$$

Next, we evaluate the orders of the three terms above.

First, we have

$$\frac{1}{N}\int \mathrm{E}\{\eta_1^2(t)\}dt = \frac{1}{N}\mathrm{E}\|\eta_0\|^2 = \mathcal{O}\left(\frac{1}{N}\right).$$

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The integral in the second term can be rewritten as

$$\int \mathrm{E}\left\{\eta_1(t)\sum_{i=1}^{N}\sum_{n=1}^{i} a^{i-n}\eta_n(t)\right\}dt = \sum_{i=1}^{N}\sum_{n=1}^{i} a^{i-n}\int \mathrm{E}\{\eta_n(t)\eta_1(t)\}dt.$$

Note that

$$\begin{aligned}\sum_{n=1}^{N}\int \mathrm{E}\{\eta_n(t)\eta_1(t)\}dt = \sum_{n=1}^{N}\int \mathrm{E}\{\eta_{n-1}(t)\eta_0(t)\}dt &= \int \sum_{n=1}^{N} \mathrm{E}\{\eta_{n-1}(t)\eta_0(t)\}dt \\ &\to \int \frac{1}{2}\{C(t,t)+\mathrm{E}\|\eta_0\|^2\}dt \\ &= \frac{1}{2}\int C(t,t)dt + \frac{1}{2}\mathrm{E}\|\eta_0\|^2 < \infty,\end{aligned} \tag{B.9}$$

when $N \to \infty$. Since $\sum_{n=1}^{\infty}\int \mathrm{E}\{\eta_n(t)\eta_1(t)\}dt < \infty$, for any $\varepsilon > 0$, there exists an integer $N_{1,\varepsilon}$ such that when $n > N_{1,\varepsilon}$, we have $|\sum_{k=n}^{\infty}\int \mathrm{E}\{\eta_k(t)\eta_1(t)\}dt| < \varepsilon$. We set

$$K_\varepsilon \triangleq \sum_{i=1}^{N_{1,\varepsilon}}\sum_{n=1}^{i} a^{i-n}\int \mathrm{E}\{\eta_n(t)\eta_1(t)\}dt \quad \text{and} \quad S_1 \triangleq \sum_{n=1}^{\infty}\int \mathrm{E}\{\eta_n(t)\eta_1(t)\}dt.$$

Then, we have

$$\sum_{i=1}^{N}\sum_{n=1}^{i} a^{i-n}\int \mathrm{E}\{\eta_n(t)\eta_1(t)\}dt = a^{N-N_{1,\varepsilon}}K_\varepsilon + \sum_{i=1}^{N-N_{1,\varepsilon}} a^{i-1}\left[\sum_{n=1}^{N-i+1}\int \mathrm{E}\{\eta_n(t)\eta_1(t)\}dt\right].$$

Since $S_1 - \varepsilon < \sum_{n=1}^{N-i+1}\int \mathrm{E}\{\eta_n(t)\eta_1(t)\}dt < S_1 + \varepsilon$ for $i = 1, \ldots, N - N_\varepsilon$, we have

$$\sum_{i=1}^{N-N_{1,\varepsilon}} a^{i-1}(S_1-\varepsilon) < \sum_{i=1}^{N-N_{1,\varepsilon}} a^{i-1}\left[\sum_{n=1}^{N-i+1}\int \mathrm{E}\{\eta_n(t)\eta_1(t)\}dt\right] < \sum_{i=1}^{N-N_{1,\varepsilon}} a^{i-1}(S_1+\varepsilon).$$

Subsequently, we have

$$\begin{aligned} a^{N-N_{1,\varepsilon}}K_\varepsilon + \frac{1-a^{N-N_{1,\varepsilon}}}{1-a}(S_1-\varepsilon) &< \sum_{i=1}^{N}\sum_{n=1}^{i} a^{i-n}\int \mathrm{E}\{\eta_n(t)\eta_1(t)\}dt \\ &< a^{N-N_{1,\varepsilon}}K_\varepsilon + \frac{1-a^{N-N_{1,\varepsilon}}}{1-a}(S_1+\varepsilon)\end{aligned}$$

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and

$$\left|\sum_{i=1}^{N}\sum_{n=1}^{i} a^{i-n}\int \mathrm{E}\{\eta_n(t)\eta_1(t)\}dt - \frac{1}{1-a}S_1\right| < a^{N-N_{1,\varepsilon}}\left|K_\varepsilon - \frac{1}{1-a}S_1\right| + \frac{1-a^{N-N_{1,\varepsilon}}}{1-a}\varepsilon. \tag{B.10}$$

By (B.10), we can find a number $M_{1,\varepsilon} > \log_a \varepsilon - \log_a \left|K_\varepsilon - \frac{1}{1-a}S_1\right| + N_{1,\varepsilon}$ such that for any $N' > M_{1,\varepsilon}$, we have

$$\left|\sum_{i=1}^{N'}\sum_{n=1}^{i} a^{i-n}\int \mathrm{E}\{\eta_n(t)\eta_1(t)\}dt - \frac{1}{1-a}S_1\right| < \frac{2-a}{1-a}\varepsilon,$$

which implies that

$$\sum_{i=1}^{N}\sum_{n=1}^{i} a^{i-n}\int \mathrm{E}\{\eta_n(t)\eta_1(t)\}dt \to \frac{1}{1-a}\sum_{n=1}^{\infty}\int \mathrm{E}\{\eta_n(t)\eta_1(t)\}dt \tag{B.11}$$

as $N \to \infty$, and conclusively,

$$\frac{2}{N^2}\int \mathrm{E}\left\{\eta_1(t)\sum_{i=1}^{N}\sum_{n=1}^{i} a^{i-n}\eta_n(t)\right\}dt = \mathcal{O}\left(\frac{1}{N^2}\right).$$

For the third term, we first rewrite it as

$$\begin{aligned}\frac{1}{N^2}\int \mathrm{E}\left[\sum_{i=1}^{N}\left\{\sum_{n=1}^{i} a^{i-n}\eta_n(t)\right\}^2\right]dt &= \frac{2}{N^2}\sum_{i=1}^{N}\sum_{n=1}^{i}\left(\sum_{k=0}^{i-n} a^{2k+n-1}\right)\int \mathrm{E}\{\eta_n(t)\eta_1(t)\}dt \\ &\quad -\frac{1}{N^2}\sum_{i=1}^{N}\sum_{k=0}^{i-1} a^{2k+n-1}\int \mathrm{E}\{\eta_1(t)\eta_1(t)\}dt\end{aligned}$$

and note that

$$\frac{1}{N^2}\sum_{i=1}^{N}\sum_{k=0}^{i-1} a^{2k+n-1}\int \mathrm{E}\{\eta_0(t)\eta_0(t)\}dt = \frac{1}{N^2}\left\{\frac{N}{1-a^2} - a^2\frac{1-a^{2N}}{(1-a^2)^2}\right\}\mathrm{E}\|\eta_0\|^2 = \mathcal{O}\left(\frac{1}{N}\right).$$

We thus focus on the remaining one term. By (B.9) and the Abel Discriminance in series, we have

$$\sum_{n=1}^{\infty} a^{n-1}\int \mathrm{E}\{\eta_n(t)\eta_1(t)\}dt < \infty.$$

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Hence, for any $\varepsilon > 0$, there exists an integer $N_{2,\varepsilon}$ such that when $n > N_{2,\varepsilon}$, we have

$$\left| \sum_{k=n}^{\infty} a^{k-1} \int \mathrm{E}\{\eta_k(t)\eta_1(t)\} dt \right| < \varepsilon.$$

We set

$$K_{1,\varepsilon} \triangleq \sum_{i=1}^{N_{2,\varepsilon}} (\sum_{k=0}^{i-1} a^{2k}) \sum_{n=1}^{N_{2,\varepsilon}-i+1} a^{n-1} \int \mathrm{E}\{\eta_n(t)\eta_1(t)\} dt,$$

$$K_{2,\varepsilon} \triangleq \sum_{i=1}^{N_{2,\varepsilon}} a^{2i-2} \sum_{n=1}^{N_{2,\varepsilon}-i+1} a^{n-1} \int \mathrm{E}\{\eta_n(t)\eta_1(t)\} dt, \quad \text{and} \quad S_2 \triangleq \sum_{n=1}^{\infty} a^{n-1} \int \mathrm{E}\{\eta_n(t)\eta_1(t)\} dt.$$

Then we obtain

$$\begin{aligned}
& \sum_{i=1}^{N} \sum_{n=1}^{i} \left(\sum_{k=0}^{i-n} a^{2k+n-1} \right) \int \mathrm{E}\{\eta_n(t)\eta_1(t)\} dt \\
= & \sum_{i=1}^{N} \left(\sum_{k=0}^{i-1} a^{2k} \right) \sum_{n=1}^{N-i+1} a^{n-1} \int \mathrm{E}\{\eta_n(t)\eta_1(t)\} dt \\
= & K_{1,\varepsilon} + \left(\sum_{k=1}^{N-N_{2,\varepsilon}} a^{2k} \right) K_{2,\varepsilon} + \sum_{i=1}^{N-N_{2,\varepsilon}} \left(\sum_{k=0}^{i-1} a^{2k} \right) \sum_{n=1}^{N-i+1} a^{n-1} \int \mathrm{E}\{\eta_n(t)\eta_1(t)\} dt.
\end{aligned}$$

Since $S_2 - \varepsilon < \sum_{n=1}^{N-i+1} a^{n-1} \int \mathrm{E}\{\eta_n(t)\eta_1(t)\} dt < S_2 + \varepsilon$ for $i = 1, \cdots, N - N_{2,\varepsilon}$, we have

$$\sum_{i=1}^{N-N_{2,\varepsilon}} \left(\sum_{k=0}^{i-1} a^{2k} \right) (S_2 - \varepsilon) < \sum_{i=1}^{N-N_{2,\varepsilon}} \left(\sum_{k=0}^{i-1} a^{2k} \right) \sum_{n=1}^{N-i+1} a^{n-1} \int \mathrm{E}\{\eta_n(t)\eta_1(t)\} dt < \sum_{i=1}^{N-N_{2,\varepsilon}} \left(\sum_{k=0}^{i-1} a^{2k} \right) (S_2 + \varepsilon).$$

Simple calculation yields that

$$\begin{aligned}
& \left\{ \frac{N - N_{2,\varepsilon}}{1 - a^2} - a^2 \frac{1 - a^{2N-2N_{2,\varepsilon}}}{(1 - a^2)^2} \right\} (S_2 - \varepsilon) \\
< & \sum_{i=1}^{N-N_{2,\varepsilon}} \left(\sum_{k=0}^{i-1} a^{2k} \right) \sum_{n=1}^{N-i+1} a^{n-1} \int \mathrm{E}\{\eta_n(t)\eta_1(t)\} dt < \left\{ \frac{N - N_{2,\varepsilon}}{1 - a^2} - a^2 \frac{1 - a^{2N-2N_{2,\varepsilon}}}{(1 - a^2)^2} \right\} (S_2 + \varepsilon).
\end{aligned}$$

Hence, we obtain that

$$\begin{aligned}
& \left| \frac{1}{N} \sum_{i=1}^{N} \left(\sum_{k=0}^{i-1} a^{2k} \right) \sum_{n=1}^{N-i+1} a^{n-1} \int \mathrm{E}\{\eta_n(t)\eta_1(t)\} dt - \frac{1}{1 - a^2} S_2 \right| \qquad \text{(B.12)} \\
< & \frac{1}{N - N_{2,\varepsilon}} |K_{1,\varepsilon}| + \frac{a^2 \left(1 - a^{2N-2N_{2,\varepsilon}}\right)}{(N - N_{2,\varepsilon})(1 - a^2)} \left| K_{2,\varepsilon} - \frac{1}{1 - a^2} S_2 \right| + \left[\frac{1}{1 - a^2} - \frac{a^2 \left(1 - a^{2N-2N_{2,\varepsilon}}\right)}{(N - N_{2,\varepsilon})(1 - a^2)^2} \right] \varepsilon.
\end{aligned}$$

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By (B.12), we can find a number $M_{2,\varepsilon} > \frac{1}{\varepsilon}\left(|K_{1,\varepsilon}| + \frac{a^2}{1-a^2}\left|K_{2,\varepsilon} - \frac{1}{1-a^2}S_2\right|\right) + N_{2,\varepsilon}$ such that for any $N' > M_{2,\varepsilon}$, we have

$$\left|\frac{1}{N'}\sum_{i=1}^{N'}\left(\sum_{k=0}^{i-1}a^{2k}\right)\sum_{n=1}^{N'-i+1}a^{n-1}\int \mathrm{E}\{\eta_n(t)\eta_1(t)\}dt - \frac{1}{1-a^2}S_2\right| < \frac{2-a^2}{1-a^2}\varepsilon,$$

which implies that

$$\frac{1}{N}\sum_{i=1}^{N}\sum_{n=1}^{i}\left(\sum_{k=0}^{i-n}a^{2k+n-1}\right)\int \mathrm{E}\{\eta_n(t)\eta_1(t)\}dt \to \frac{1}{1-a^2}S_2$$

as $N \to \infty$. Then it is easy to see that

$$\frac{2}{N^2}\sum_{i=1}^{N}\sum_{n=1}^{i}(\sum_{k=0}^{i-n}a^{2k+n-1})\int \mathrm{E}\{\eta_n(t)\eta_1(t)\}dt = \mathcal{O}\left(\frac{1}{N}\right)$$

and

$$\frac{1}{N^2}\int \mathrm{E}\left\{\sum_{i=1}^{N}\left\{\sum_{n=1}^{i}a^{i-n}\eta_n(t)\right\}^2\right\}dt = \mathcal{O}\left(\frac{1}{N}\right).$$

To sum up, under the alternative hypothesis H_1^{ar1}, $\mathrm{E}(R_N) = \mathcal{O}(1/N)$, which implies $R_N = \mathcal{O}_{\mathrm{Pr}}(1/N)$. □

Appendix B.6. Proof of Theorem 6

Proof of Theorem 6. We mainly investigate the order of the estimate of the long-run covariance function $\hat{c}_N(t,s)$. Different from the proof of Theorem 6, there is not a term of "trend" in the estimated errors that will dominate the order of the long-run covariance function. We will however apply some results in Horváth et al. (2013).

Under the alternative hypothesis H_1^{ws}, we have

$$\hat{\eta}_n(t) = X_{n+1}(t) - X_n(t) = \eta_{n+1}(t) - \eta_n(t).$$

Let us consider a general form of long-run covariance function in (3.10). By Theorems 1 and 2 in Horváth et al. (2013), under some Conditions (*) specified below, we will have the following two claims:

1. The infinite sums

$$\bar{c}(t,s) = \mathrm{E}\{\hat{\eta}_0(t)\hat{\eta}_0(s)\} + \sum_{n\geq 1}\mathrm{E}\{\hat{\eta}_0(t)\hat{\eta}_n(s)\} + \sum_{n\geq 1}\mathrm{E}\{\hat{\eta}_0(s)\hat{\eta}_n(t)\} \quad \text{(B.13)}$$

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converge in $L^2([0,1]\times[0,1])$, i.e. $\bar{c}(t,s)$ is square integrable function on the unit square and it is easy to derive that such a square integration is positive.

2. Moreover,

$$\iint \{\hat{c}_N(t,s) - \bar{c}(t,s)\}^2 dtds \xrightarrow{p} 0 \quad as \quad N \to \infty. \tag{B.14}$$

These two claims assert that $\hat{c}_N(t,s)$ will converges in $L^2([0,1]\times[0,1])$ in probability and also have a positive square integration independent of N. Together with Dunford and Schwartz (1988, p.1091), the proof is completed. It suffices for us to show the Conditions (*) hold for our application.

Conditions (*) refer to (1.1), (1.2), (2.1), (2.2), (2.7)-(2.10), (2.13) and (2.14) in Horváth et al. (2013). Since some of these conditions have been nested within the specification and assumption of the errors in this paper, we only need to prove the following five conditions:

(c1) Estimated errors $\hat{\eta}_n$ can be write as $\hat{\eta}_n = F(\varepsilon_n, \varepsilon_{n-1}, \ldots)$ for some measurable function $F : \Omega^\infty \to L^2$ and i.i.d. random functions $\varepsilon_n, -\infty < n < \infty$.

(c2) $\mathrm{E}(\hat{\eta}_0) = 0$ in L^2.

(c3) $\int \mathrm{E}\{\hat{\eta}_0^2(t)\}dt < \infty$.

(c4) $\sum_{m\geq 1}[\mathrm{E}\int\{\hat{\eta}_n(t) - \hat{\eta}_{n,m}(t)\}^2 dt]^{1/2} < \infty$, where $\hat{\eta}_{n,m} = F(\varepsilon_n, \ldots, \varepsilon_{n-m+1}, \varepsilon^{(m)}_{n,n-m}, \varepsilon^{(m)}_{n,n-m-1}, \ldots)$ and $\{\varepsilon^{(m)}_{n,k}\}_{k=-\infty}^{n-m}$ are independent copies of ε_n.

(c5) $\lim_{m\to\infty} m[\mathrm{E}\int\{\hat{\eta}_n(t) - \hat{\eta}_{n,m}(t)\}^2 dt]^{1/2} = 0$, where $\hat{\eta}_{n,m}$ is the same as mentioned in (c4).

For **(c1)**, since $\hat{\eta}_n = \eta_{n+1} - \eta_n = f(\varepsilon_{n+1}, \varepsilon_n, \ldots) - f(\varepsilon_n, \varepsilon_{n-1}, \ldots)$, f is a measurable function, and $\varepsilon_{n+1}, \varepsilon_n, \ldots$ are i.i.d. random functions, we can set $F(\varepsilon_{n+1}, \varepsilon_n, \ldots) \triangleq f(\varepsilon_{n+1}, \varepsilon_n, \ldots) - f(\varepsilon_n, \varepsilon_{n-1}, \ldots) = \hat{\eta}_n$, then F is a measurable function.

For **(c2)** and **(c3)**, it is clear to that

$$\mathrm{E}(\hat{\eta}_0) = \mathrm{E}(\eta_1 - \eta_0) = \mathrm{E}(\eta_1) - \mathrm{E}(\eta_0) = 0 \quad in \quad L^2$$

and

$$\int \mathrm{E}\{\hat{\eta}_0^2(t)\}dt = \int \mathrm{E}\{\eta_1(t) - \eta_0(t)\}^2 dt \leq \int 2\mathrm{E}\{\eta_1^2(t) + \eta_0^2(t)\}dt < \infty$$

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For **(c4)**, we define

$$\begin{aligned}\hat{\eta}_{n,m} &\triangleq F(\varepsilon_{n+1},\ldots,\varepsilon_{n-m+2},\varepsilon^{(m)}_{n+1,n-m+1},\varepsilon^{(m)}_{n+1,n-m},\ldots)\\ &= f(\varepsilon_{n+1},\ldots,\varepsilon_{n-m+2},\varepsilon^{(m)}_{n+1,n-m+1},\ldots)-f(\varepsilon_{n},\ldots,\varepsilon_{n-m+2},\varepsilon^{(m)}_{n+1,n-m+1},\ldots)\\ &\triangleq \tilde{\eta}_{n+1,m}-\tilde{\eta}_{n,m-1}, \qquad \text{(B.15)}\end{aligned}$$

where $\{\varepsilon^{(m)}_{n+1,k}\}_{k=-\infty}^{n-m+1}$ are independent copies of ε_{n+1}. We do not use $\eta_{n+1,m}$ and $\eta_{n,m-1}$ to represent the two functions of f because the correlation between $\tilde{\eta}_{n+1,m}$ and $\tilde{\eta}_{n,m-1}$ is different from $\eta_{n+1,m}$ and $\eta_{n,m-1}$. Despite of this, $\tilde{\eta}_{n+1,m}$ and $\eta_{n+1,m}$, $\tilde{\eta}_{n,m-1}$ and $\eta_{n,m-1}$ share the same distribution. Then, we have

$$\begin{aligned}&\left[\mathrm{E}\int\{\hat{\eta}_n(t)-\hat{\eta}_{n,m}(t)\}^2dt\right]^{1/2}\\ =\;&\left[\mathrm{E}\int[\{\eta_{n+1}(t)-\eta_n(t)\}-\{\tilde{\eta}_{n+1,m}(t)-\tilde{\eta}_{n,m-1}(t)\}]^2dt\right]^{1/2}\\ =\;&\left[\mathrm{E}\int[\{\eta_{n+1}(t)-\tilde{\eta}_{n+1,m}(t)\}-\{\eta_n(t)-\tilde{\eta}_{n,m-1}(t)\}]^2dt\right]^{1/2}\\ \le\;&\left[2\mathrm{E}\int\{\eta_{n+1}(t)-\tilde{\eta}_{n+1,m}(t)\}^2dt+2\mathrm{E}\int\{\eta_n(t)-\tilde{\eta}_{n,m-1}(t)\}^2dt\right]^{1/2}\\ =\;&\sqrt{2}\left[\mathrm{E}\int\{\eta_{n+1}(t)-\eta_{n+1,m}(t)\}^2dt+\mathrm{E}\int\{\eta_n(t)-\eta_{n,m-1}(t)\}^2dt\right]^{1/2}\\ \le\;&\sqrt{2}\left[\mathrm{E}\int\{\eta_{n+1}(t)-\eta_{n+1,m}(t)\}^2dt\right]^{1/2}+\sqrt{2}\left[\mathrm{E}\int\{\eta_n(t)-\eta_{n,m-1}(t)\}^2dt\right]^{1/2}, \qquad \text{(B.16)}\end{aligned}$$

which implies that

$$\begin{aligned}&\sum_{m\ge1}\left[\mathrm{E}\int\{\hat{\eta}_n(t)-\hat{\eta}_{n,m}(t)\}\right]^{1/2}\\ \le\;&\sqrt{2}\sum_{m\ge1}\left[\mathrm{E}\int\{\eta_{n+1}(t)-\eta_{n+1,m}(t)\}^2dt\right]^{1/2}+\sqrt{2}\sum_{m\ge1}\left[\mathrm{E}\int\{\eta_n(t)-\eta_{n,m-1}(t)\}^2dt\right]^{1/2}\\ \le\;&\sqrt{2}\sum_{m\ge1}\left[\mathrm{E}\int\{\eta_{n+1}(t)-\eta_{n+1,m}(t)\}^2dt\right]^{1/2}+\sqrt{2}\sum_{m\ge1}\left[\mathrm{E}\int\{\eta_n(t)-\eta_{n,m}(t)\}^2dt\right]^{1/2}\\ &+\sqrt{2}\left[\mathrm{E}\int\{\eta_n(t)-\eta_{n,0}(t)\}^2dt\right]^{1/2}.\end{aligned}$$

The first two terms are finite by the assumption of the errors. Thus, condition **(c4)** is proved.

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For **(c5)**, we can obtain from (B.16) that

$$m\left[\mathrm{E}\int\{\hat{\eta}_n(t)-\hat{\eta}_{n,m}(t)\}\right]^{1/2} \leq \sqrt{2}m\left[\mathrm{E}\int\{\eta_{n+1}(t)-\eta_{n+1,m}(t)\}^2dt\right]^{1/2} + \sqrt{2}m\left[\mathrm{E}\int\{\eta_n(t)-\eta_{n,m-1}(t)\}^2dt\right]^{1/2}.$$

It is clear that the right-hand side of the inequality will tend to 0 as $m\to\infty$ by the assumption (2.2). Thus, condition **(c5)** is proved. □

Appendix B.7. Proof of Theorem 7

Proof of Theorem 7. Similar to the proof of Theorem 6, it suffices to show the conditions **(c1)**-**(c5)** in Section Appendix B.6 hold. The only difference is that the estimated errors now become

$$\hat{\eta}_n(t)=X_{n+1}(t)-X_n(t)=\sum_{k=0}^{\infty}a^k\eta_{n-k+1}(t)-\sum_{k=0}^{\infty}a^k\eta_{n-k}(t)=\eta_{n+1}(t)+\sum_{k=0}^{\infty}(a^{k+1}-a^k)\eta_{n-k}(t).$$

Here, we assume the data set is $\{X_n\}_{n=-\infty}^{\infty}$ instead of starting from X_1 or X_0. This consideration will better explain the generalization of the estimated errors under H_1^{ar1}.

For **(c1)**, since $\hat{\eta}_n(t)=X_{n+1}(t)-X_n(t)$ is the difference of two continuous and bounded functional data, $\hat{\eta}_n(t)$ itself is a measurable function. We can then set

$$\begin{aligned}F(\varepsilon_{n+1},\varepsilon_n,\ldots) &\triangleq \sum_{k=0}^{\infty}a^kf(\varepsilon_{n-k+1},\varepsilon_{n-k},\ldots)-\sum_{k=0}^{\infty}a^kf(\varepsilon_{n-k},\varepsilon_{n-k-1},\ldots)\\ &= f(\varepsilon_{n+1},\varepsilon_n,\ldots)+\sum_{k=0}^{\infty}(a^{k+1}-a^k)f(\varepsilon_{n-k},\varepsilon_{n-k-1},\ldots)=\hat{\eta}_n,\end{aligned}$$

where F is a measurable function.

For **(c2)** and **(c3)**, we note that

$$\mathrm{E}(\hat{\eta}_0)=\mathrm{E}(\eta_{n+1})+\sum_{k=0}^{\infty}(a^{k+1}-a^k)\mathrm{E}(\eta_{n-k})=0,$$

and

$$\begin{aligned}\int\mathrm{E}\{\eta_0^2(t)\}dt &= \int\mathrm{E}\{\eta_{n+1}(t)+\sum_{k=0}^{\infty}(a^{k+1}-a^k)\eta_{n-k}(t)\}^2dt\\ &\leq 2\int\mathrm{E}\{\eta_{n+1}^2(t)+\sum_{k=0}^{\infty}(a^{k+1}-a^k)\eta_{n-k}^2(t)\}dt<\infty.\end{aligned}$$

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For **(c4)**, let us define

$$\begin{aligned}
\hat{\eta}_{n,m} &\triangleq F(\varepsilon_{n+1},\ldots,\varepsilon_{n-m+2},\varepsilon^{(m)}_{n+1,n-m+1},\varepsilon^{(m)}_{n+1,n-m},\ldots)\\
&= f(\varepsilon_{n+1},\ldots,\varepsilon_{n-m+2},\varepsilon^{(m)}_{n+1,n-m+1},\ldots)\\
&\quad+\sum_{k=0}^{m-2}(a^{k+1}-a^k)f(\varepsilon_{n-k},\ldots,\varepsilon_{n-m+2},\varepsilon^{(m)}_{n+1,n-m+1},\ldots)\\
&\quad+\sum_{k=m-1}^{\infty}(a^{k+1}-a^k)f(\varepsilon^{(m)}_{n+1,n-k},\varepsilon^{(m)}_{n+1,n-k-1},\ldots)\\
&= \tilde{\eta}_{n+1,m}+\sum_{k=0}^{m-2}(a^{k+1}-a^k)\tilde{\eta}_{n-k,m-k-1}+\sum_{k=m-1}^{\infty}(a^{k+1}-a^k)\tilde{\eta}_{n-k,0},
\end{aligned}$$

where $\{\eta^{(m)}_{n+1,k}\}_{k=-\infty}^{n-m+1}$ are independent copies of η_{n+1}. Similar to the discussion of (B.15), $\tilde{\eta}_{p,q}$ has the same distribution as $\eta_{p,q}$ but the correlations may be different. Then, we have

$$\begin{aligned}
&\mathrm{E}\int\{\hat{\eta}_n(t)-\hat{\eta}_{n,m}(t)\}^2dt\\
=\;&\mathrm{E}\int\Bigg[\{\eta_{n+1}(t)+\sum_{k=0}^{\infty}(a^{k+1}-a^k)\eta_{n-k}(t)\}-\{\tilde{\eta}_{n+1,m}(t)\\
&\quad+\sum_{k=0}^{m-2}(a^{k+1}-a^k)\tilde{\eta}_{n-k,m-k-1}(t)+\sum_{k=m-1}^{\infty}(a^{k+1}-a^k)\tilde{\eta}_{n-k,0}(t)\}\Bigg]^2dt\\
=\;&\mathrm{E}\int\Bigg[\{\eta_{n+1}(t)-\tilde{\eta}_{n+1,m}(t)\}+\sum_{k=0}^{m-2}(a^{k+1}-a^k)\{\eta_{n-k}(t)-\tilde{\eta}_{n-k,m-k-1}(t)\}\\
&\quad+\sum_{k=m-1}^{\infty}(a^{k+1}-a^k)\{\eta_{n-k}(t)-\tilde{\eta}_{n-k,0}(t)\}\Bigg]^2dt\\
\le\;&2\mathrm{E}\int\{\eta_{n+1}(t)-\tilde{\eta}_{n+1,m}(t)\}^2dt+2\sum_{k=0}^{m-2}(a^k-a^{k+1})\mathrm{E}\int\{\eta_{n-k}(t)-\tilde{\eta}_{n-k,m-k-1}(t)\}^2dt\\
&\quad+2\sum_{k=m-1}^{\infty}(a^k-a^{k+1})\mathrm{E}\int\{\eta_{n-k}(t)-\tilde{\eta}_{n-k,0}(t)\}^2dt.\\
=\;&2\mathrm{E}\int\{\eta_{n+1}(t)-\eta_{n+1,m}(t)\}^2dt+2\sum_{k=0}^{m-2}(a^k-a^{k+1})\mathrm{E}\int\{\eta_{n-k}(t)-\eta_{n-k,m-k-1}(t)\}^2dt\\
&\quad+2\sum_{k=m-1}^{\infty}(a^k-a^{k+1})\mathrm{E}\int\{\eta_{n-k}(t)-\eta_{n-k,0}(t)\}^2dt.
\end{aligned}$$

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Since η_{n-k} and $\eta_{n-k,0}$ are independent, we have $\mathrm{E}\int\{\eta_{n-k}(t)-\eta_{n-k,0}(t)\}^2dt = 2\mathrm{E}\int\eta^2_{n-k}(t)dt = 2\mathrm{E}\int\eta_0^2(t)dt$, which is a fixed value and denoted as σ^2. Then we have

$$\begin{aligned}
&\sum_{m\geq 1}\left[\mathrm{E}\int\{\hat{\eta}_n(t)-\hat{\eta}_{n,m}(t)\}^2dt\right]^{1/2}\\
\leq\ &\sum_{m\geq 1}\sqrt{2}\left[\mathrm{E}\int\{\eta_{n+1}(t)-\eta_{n+1,m}(t)\}^2dt\right.\\
&\left.+\sum_{k=0}^{m-2}(a^k-a^{k+1})\mathrm{E}\int\{\eta_{n-k}(t)-\eta_{n-k,m-k-1}(t)\}^2dt+a^{m-1}\sigma^2\right]^{1/2}\\
\leq\ &\sqrt{2}\sum_{m\geq 1}\left[\mathrm{E}\int\{\eta_{n+1}(t)-\eta_{n+1,m}(t)\}^2dt\right]^{1/2}\\
&+\sqrt{2}\sum_{m\geq 2}\sum_{k=0}^{m-2}\sqrt{a^k-a^{k+1}}\left[\mathrm{E}\int\{\eta_{n-k}(t)-\eta_{n-k,m-k-1}(t)\}^2dt\right]^{1/2}\\
&+\sqrt{2}\sum_{m\geq 2}\sqrt{a^{m-1}}\sigma^2 \qquad\qquad (B.17)
\end{aligned}$$

In (B.17), the first and the third terms are clearly finite by the assumptions of this paper. Let us denote that $T \triangleq \sum_{m\geq 1}\left[\mathrm{E}\int\{\eta_0(t)-\eta_{0,m}(t)\}^2dt\right]^{1/2}$, then for the second term above, we have

$$\begin{aligned}
&\sum_{m\geq 2}\sum_{k=0}^{m-2}\sqrt{a^k-a^{k+1}}\left[\mathrm{E}\int\{\eta_{n-k}(t)-\eta_{n-k,m-k-1}(t)\}^2dt\right]^{1/2}\\
=\ &\sum_{k=0}^{\infty}\sqrt{a^k-a^{k+1}}\sum_{m\geq k+2}\left[\mathrm{E}\int\{\eta_{n-k}(t)-\eta_{n-k,m-k-1}(t)\}^2dt\right]^{1/2}\\
=\ &\sum_{k=0}^{\infty}\sqrt{a^k-a^{k+1}}\sum_{m\geq 1}\left[\mathrm{E}\int\{\eta_{n-k}(t)-\eta_{n-k,m}(t)\}^2dt\right]^{1/2}\\
=\ &\sum_{k=0}^{\infty}\sqrt{a^k-a^{k+1}}T=\frac{\sqrt{1-a}}{1-\sqrt{a}}T<\infty.
\end{aligned}$$

Thus, condition **(c4)** is proved.

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For **(c5)**, we have

$$
\begin{aligned}
& m\left[\mathrm{E}\int\{\hat{\eta}_n(t)-\hat{\eta}_{n,m}(t)\}^2dt\right]^{1/2} \\
\leq & \sqrt{2}m\left[\mathrm{E}\int\{\eta_{n+1}(t)-\eta_{n+1,m}(t)\}^2dt\right. \\
& \left.+\sum_{k=0}^{m-2}(a^k-a^{k+1})\mathrm{E}\int\{\eta_{n-k}(t)-\eta_{n-k,m-k-1}(t)\}^2dt+a^{m-1}\sigma^2\right]^{1/2} \\
\leq & \sqrt{2}m\left[\mathrm{E}\int\{\eta_{n+1}(t)-\eta_{n+1,m}(t)\}^2dt\right]^{1/2} \\
& +\sqrt{2}m\left[\sum_{k=0}^{m-2}(a^k-a^{k+1})\mathrm{E}\int\{\eta_{n-k}(t)-\eta_{n-k,m-k-1}(t)\}^2dt\right]^{1/2}+\sqrt{2}m\sqrt{a^{m-1}}\sigma^2.
\end{aligned}
\tag{B.18}
$$

In (B.18), it is clear that the first and the third terms tend to 0 when $m\to\infty$ by the assumptions of this paper. We define $K_m \triangleq [\mathrm{E}\int\{\eta_0(t)-\eta_{0,m}\}^2dt]^{1/2}$. By the assumption (2), we have $\lim_{m\to\infty} mK_m = 0$, i.e., for any $\varepsilon>0$, there exists an integer $N_\varepsilon>0$ such that for any $m>N_\varepsilon$, we have $K_m<\frac{\varepsilon}{m}$. Then we consider the second term in (B.18), we have

$$
\begin{aligned}
& m\left[\sum_{k=0}^{m-2}(a^k-a^{k+1})\mathrm{E}\int\{\eta_{n-k}(t)-\eta_{n-k,m-k-1}(t)\}^2dt\right]^{1/2} \\
= & \left[m^2\sum_{k=0}^{m-2}(a^k-a^{k+1})K_{m-k-1}^2\right]^{1/2}=\sqrt{1-a}\left[m^2\sum_{k=0}^{m-2}a^kK_{m-k-1}^2\right]^{1/2} \\
= & \sqrt{1-a}\left[m^2\sum_{k=m-N_\varepsilon-1}^{m-2}a^kK_{m-k-1}^2+m^2\sum_{k=0}^{m-N_\varepsilon-2}a^kK_{m-k-1}^2\right]^{1/2}. \qquad \text{(B.19)}
\end{aligned}
$$

In (B.19), for the first term,

$$
m^2\sum_{k=m-N_\varepsilon-1}^{m-2}a^kK_{m-k-1}^2=m^2a^{m-N_\varepsilon-1}\sum_{k=1}^{N_\varepsilon}a^{N_\varepsilon-k}K_k^2\to 0
$$

when $m\to\infty$, and for the second term,

$$
\begin{aligned}
m^2\sum_{k=0}^{m-N_\varepsilon-2}a^kK_{m-k-1}^2 & < m^2\sum_{k=0}^{m-N_\varepsilon-2}a^k\frac{\varepsilon^2}{(m-k-1)^2}<m^2\sum_{k=0}^{m-2}a^k\frac{\varepsilon^2}{(m-k-1)^2} \\
& = \epsilon^2m^2\sum_{k=1}^{m-1}\frac{a^{k-1}}{(m-k)^2}\triangleq\epsilon^2S_m.
\end{aligned}
$$

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Then we have

$$\begin{aligned} S_m - aS_m &= m^2 \left\{ \sum_{k=1}^{m-1} \frac{a^{k-1}}{(m-k)^2} - \sum_{k=1}^{m-1} \frac{a^k}{(m-k)^2} \right\} \\ &= \left\{ \frac{m^2}{(m-1)^2} - m^2 a^{m-1} \right\} + m^2 \sum_{k=1}^{m-2} a^k \left\{ \frac{1}{(m-k-1)^2} - \frac{1}{(m-k)^2} \right\} \\ &= \left\{ \frac{m^2}{(m-1)^2} - m^2 a^{m-1} \right\} + \sum_{k=1}^{m-2} a^k \frac{m^2(2m+2k-1)}{(m-k-1)^2(m-k)^2}. \end{aligned} \tag{B.20}$$

The first term in (B.20) tends to 1 as $m \to \infty$. For the second term in (B.20), we define $m^* \triangleq [\sqrt{m}]$. Then we have

$$\sum_{k=1}^{m^*} a^k \frac{m^2(2m+2k-1)}{(m-k-1)^2(m-k)^2} \leq \sum_{k=1}^{m^*} \mathcal{O}\left(\frac{1}{m}\right) = \mathcal{O}\left(\frac{1}{\sqrt{m}}\right)$$

and

$$\sum_{k=m^*+1}^{m-1} a^k \frac{m^2(2m+2k-1)}{(m-k-1)^2(m-k)^2} \leq a^{\sqrt{m}} \frac{m^3(4m-5)}{4} \to 0$$

as $m \to \infty$. Hence, the second term in (B.20) also tends to 0 and thus S_m tends to $\frac{1}{1-a}$ when $m \to \infty$. Subsequently, the second term in (B.19) yields that

$$m^2 \sum_{k=0}^{m-N_\varepsilon-2} a^k K_{m-k-1}^2 < \varepsilon^2 S_m \to \frac{\varepsilon^2}{1-a}$$

as $m \to \infty$. Because we can choose any $\varepsilon > 0$, we have $m^2 \sum_{k=0}^{m-N_\varepsilon-2} a^k K_{m-k-1}^2 \to 0$ as $m \to \infty$. Then we can conclude that the second term in (B.18) tends to 0 as $m \to \infty$, which implies that

$$\lim_{m\to\infty} m \left[\mathrm{E} \int \{\hat{\eta}_n(t) - \hat{\eta}_{n,m}(t)\}^2 dt \right]^{1/2} = 0.$$

Thus, condition **(c5)** is proved. □

Appendix C. Simulation Results: Empirical Sizes and Powers of the Functional Stationarity Test

In this appendix, we simply present the result tables for the FSTs with different statistics and $h(N)$ choices, while their comparison with FURTs is contained in the main text. Note that for each setting, only left- or right-tailed test is presented as the another one is obviously wrong with nearly no power. The tables below are clustered as follows:

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- Tables C.5 and C.6 use statistic T_N and $h = N^{2/5}$ (best results).
- Tables C.7 and C.8 use statistic M_N and $h = N^{2/5}$.
- Tables C.9 and C.10 use statistic T_N and $h = N^{1/2}$.
- Tables C.11 and C.12 use statistic M_N and $h = N^{1/2}$.

Table C.5:
Empirical sizes (%) and powers (%) for (right-tailed) FST (T_N) for H_0^{ur} versus H_1^{ts} with the bandwidth of $h = N^{2/5}$.

Error type	i.i.d. Normal				AR(1) Normal			
N	100		200		100		200	
α	5%	10%	5%	10%	5%	10%	5%	10%
H_0^{ur}	3.0	6.0	2.3	7.3	2.4	6.3	3.6	9.1
H_1^{ts}	87.0	89.0	94.0	95.0	82.5	86.1	88.0	90.0
Error type	i.i.d. Student's t_5				AR(1) Student's t_5			
N	100		200		100		200	
α	5%	10%	5%	10%	5%	10%	5%	10%
H_0^{ur}	5.3	8.7	1.7	5.7	5.0	9.0	5.0	10.7
H_1^{ts}	91.0	92.3	90.7	92.0	86.3	89.7	84.7	86.0

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Table C.6:
Empirical sizes (%) and powers (%) for (left-tailed) FST (T_N) for H_0^{ur} versus H_1^{ws} or H_1^{ar1} with the bandwidth of $h = N^{2/5}$.

Error type	i.i.d. Normal				AR(1) Normal			
N	100		200		100		200	
α	5%	10%	5%	10%	5%	10%	5%	10%
H_0^{ur}	1.3	5.3	2.3	5.3	1.2	5.9	3.2	6.0
H_1^{ws}	////////////////		////////////////		5.4	22.7	27.7	58.6
H_1^{ar1} (const. kernel)	10.7	27.0	54.0	73.0	11.5	31.9	61.0	76.8
H_1^{ar1} (exp. kernel)	3.7	17.3	11.3	36.7	27.7	49.8	66.5	85.9
Error type	i.i.d. Student's t_5				AR(1) Student's t_5			
N	100		200		100		200	
α	5%	10%	5%	10%	5%	10%	5%	10%
H_0^{ur}	0.0	3.7	1.7	5.7	0.3	4.0	1.7	6.0
H_1^{ws}	////////////////		////////////////		8.3	30.3	32.7	59.0
H_1^{ar1} (const. kernel)	10.3	29.3	52.3	92.3	13.3	21.3	58.7	78.3
H_1^{ar1} (exp. kernel)	5.0	21.3	16.0	42.0	8.3	10.3	32.7	59.0

Table C.7:
Empirical sizes (%) and powers (%) for (right-tailed) FST (M_N) for H_0^{ur} versus H_1^{ts} with the bandwidth of $h = N^{2/5}$.

Error type	i.i.d. Normal				AR(1) Normal			
N	100		200		100		200	
α	5%	10%	5%	10%	5%	10%	5%	10%
H_0^{ur}	2.3	7.3	2.3	8.7	1.6	7.1	2.8	9.5
H_1^{ts}	86.7	89.0	94.7	96.7	82.9	84.5	85.7	88.4
Error type	i.i.d. Student's t_5				AR(1) Student's t_5			
N	100		200		100		200	
α	5%	10%	5%	10%	5%	10%	5%	10%
H_0^{ur}	1.3	4.3	4.3	8.7	1.0	5.0	4.7	12.0
H_1^{ts}	89.7	91.3	92.7	94.0	86.0	88.3	82.0	84.3

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Table C.8:
Empirical sizes (%) and powers (%) for (left-tailed) FST (M_N) for H_0^{ur} versus H_1^{ws} or H_1^{ar1} with the bandwidth of $h = N^{2/5}$.

Error type	i.i.d. Normal				AR(1) Normal			
N	100		200		100		200	
α	5%	10%	5%	10%	5%	10%	5%	10%
H_0^{ur}	0.3	1.3	0.3	4.7	0.4	2.0	1.2	3.6
H_1^{ws}	////////////////		////////////////		0.0	4.7	5.5	29.3
H_1^{ar1} (const. kernel)	1.0	7.0	31.3	59.7	1.9	9.6	39.8	69.1
H_1^{ar1} (exp. kernel)	0.0	3.0	2.3	13.0	3.5	24.6	52.1	83.5
Error type	i.i.d. Student's t_5				AR(1) Student's t_5			
N	100		200		100		200	
α	5%	10%	5%	10%	5%	10%	5%	10%
H_0^{ur}	0.0	0.3	1.3	3.7	0.0	0.0	2.0	3.7
H_1^{ws}	////////////////		////////////////		0.0	6.3	4.3	28.7
H_1^{ar1} (const. kernel)	1.3	10.0	32.7	61.0	3.0	14.3	42.7	67.3
H_1^{ar1} (exp. kernel)	0.0	2.3	1.0	15.3	0.0	6.3	4.3	28.7

Table C.9:
Empirical sizes (%) and powers (%) for (right-tailed) FST (T_N) for H_0^{ur} versus H_1^{ts}, H_1^{ws} or H_1^{ar1}(exp. kernel) with the bandwidth of $h = N^{1/2}$.

Error type	i.i.d. Normal				AR(1) Normal			
N	100		200		100		200	
α	5%	10%	5%	10%	5%	10%	5%	10%
H_0^{ur}	2.0	7.7	2.7	9.3	1.9	7.3	2.7	10.1
H_1^{ts}	98.7	99.7	99.6	99.7	89.9	92.2	95.7	96.1
H_1^{ws}	////////////////		////////////////		1.1	3.0	0.7	6.0
H_1^{ar1} (exp. kernel)	1.7	4.0	0.0	1.0	0.0	0.0	0.0	0.0
Error type	i.i.d. Student's t_5				AR(1) Student's t_5			
N	100		200		100		200	
α	5%	10%	5%	10%	5%	10%	5%	10%
H_0^{ur}	3.3	9.0	4.0	11.3	3.6	9.6	3.6	11.7
H_1^{ts}	98.7	99.3	98.3	98.7	94.0	95.6	91.9	92.3
H_1^{ws}	////////////////		////////////////		0.0	1.7	1.3	4.3
H_1^{ar1} (exp. kernel)	0.3	3.3	0.4	0.6	0.0	0.0	0.0	0.0

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Table C.10:
Empirical sizes (%) and powers (%) for (left-tailed) FSTs(T_N) for H_0^{ur} versus H_1^{ar1}(const. kernel) with the bandwidth of $h = N^{1/2}$.

Error type	i.i.d. Normal				AR(1) Normal			
N	100		200		100		200	
α	5%	10%	5%	10%	5%	10%	5%	10%
H_0^{ur}	0.0	0.7	1.0	3.7	0.4	1.5	1.2	3.5
H_1^{ar1} (const. kernel)	0.3	2.3	14.0	42.7	0.7	7.5	21.7	52.8
Error type	i.i.d. Student's t_5				AR(1) Student's t_5			
N	100		200		100		200	
α	5%	10%	5%	10%	5%	10%	5%	10%
H_0^{ur}	0.0	0.3	0.3	2.0	0.0	0.8	0.4	2.4
H_1^{ar1} (const. kernel)	0.0	4.3	12.0	35.7	0.8	10.8	18.6	45.7

Table C.11:
Empirical sizes (%) and powers (%) for (right-tailed) FST (M_N) for H_0^{ur} versus H_1^{ts}, H_1^{ws} or H_1^{ar1}(exp. kernel) with the bandwidth of $h = N^{1/2}$.

Error type	i.i.d. Normal				AR(1) Normal			
N	100		200		100		200	
α	5%	10%	5%	10%	5%	10%	5%	10%
H_0^{ur}	1.3	5.7	1.3	6.7	0.4	5.4	1.7	5.8
H_1^{ts}	99.3	99.7	100.0	100.0	90.3	93.4	95.3	96.1
H_1^{ws}	////////////////////		////////////////////		8.7	20.4	18.9	35.5
H_1^{ar1} (exp. kernel)	9.3	19.3	3.3	8.3	0.0	0.2	0.0	0.3
Error type	i.i.d. Student's t_5				AR(1) Student's t_5			
N	100		200		100		200	
α	5%	10%	5%	10%	5%	10%	5%	10%
H_0^{ur}	0.0	3.3	1.3	10.7	0.0	2.8	0.1	10.1
H_1^{ts}	100.0	100.0	99.7	100.0	94.0	96.4	93.1	95.5
H_1^{ws}	////////////////////		////////////////////		7.7	14.3	19.0	33.3
H_1^{ar1} (exp. kernel)	7.3	18.7	2.3	10.6	0.0	0.0	0.0	0.4

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Table C.12:
Empirical sizes (%) and powers (%) for (left-tailed) FST (M_N) for H_0^{ur} versus H_1^{ar1}(const. kernel) with the bandwidth of $h = N^{1/2}$.

Error type	i.i.d. Normal				AR(1) Normal			
N	100		200		100		200	
α	5%	10%	5%	10%	5%	10%	5%	10%
H_0^{ur}	0.0	0.0	0.0	0.7	0.0	0.0	0.0	0.4
H_1^{ar1} (const. kernel)	0.0	0.0	0.3	7.7	0.0	0.4	1.9	15.4
Error type	i.i.d. Student's t_5				AR(1) Student's t_5			
N	100		200		100		200	
α	5%	10%	5%	10%	5%	10%	5%	10%
H_0^{ur}	0.0	0.0	0.0	1.0	0.0	0.0	0.0	1.2
H_1^{ar1} (const. kernel)	0.0	0.3	0.7	6.0	0.0	0.7	1.2	12.0

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